

Chapter 2

Binomial Theorem

Solutions

SECTION - A

Objective Type Questions (One option is correct)

[Basic of Binomial Theorem for Positive Index]

1. The expansion $(a + x)^n = {}^nC_0 a^n + {}^nC_1 a^{n-1}x + \dots + {}^nC_n x^n$ is valid when n is
(1) An integer (2) A rational number (3) An irrational number (4) A natural number

Sol. Answer (4)

The given expansion is valid for all positive integral values of n (natural number).

2. If the coefficient of r th term and $(r + 1)$ th term in the expansion of $(1 + x)^{20}$ are in the ratio 1 : 2, then r is equal to
(1) 6 (2) 7 (3) 8 (4) 9

Sol. Answer (2)

$$\frac{{}^{20}C_{r-1}}{{}^{20}C_{r-0}} = \frac{1}{2}$$

$$\Rightarrow r = 7$$

3. When n is any positive integer, the expansion $(x + a)^n = {}^nC_0 x^n + {}^nC_1 x^{n-1}a + \dots + {}^nC_n a^n$ is valid only when
(1) $|x| < 1$ (2) $|x| > 1$
(3) $|x| < 1$ and $|a| < 1$ (4) x and a are any two numbers

Sol. Answer (4)

The given expansion is valid whatever x and a may be.

4. The 6th term in the expansion of $\left(x - \frac{1}{x}\right)^{10}$ is

- (1) ${}^{10}C_6 x^6$ (2) ${}^{10}C_5$ (3) $-({}^{10}C_5)$ (4) $-({}^{10}C_6 x^6)$

Sol. Answer (3)

$$T_6 = T_{5+1} = {}^{10}C_5 (x)^{(10-5)} \left(-\frac{1}{x}\right)^5 = -({}^{10}C_5)$$

5. The exponent of x occurring in the 7th term of the expansion of $\left(\frac{ax}{2} - \frac{8}{bx}\right)^9$ is

(1) 3

(2) -3

(3) 5

(4) -5

Sol. Answer (2)

The general term

$$T_{r+1} = {}^9C_r \left(\frac{ax}{2}\right)^{9-r} \left(\frac{-8}{bx}\right)^r$$

$$T_7 = {}^9C_6 \left(\frac{ax}{2}\right)^{9-6} \left(\frac{-8}{bx}\right)^6$$

$$= {}^9C_6 \left(\frac{a}{2}\right)^3 \left(\frac{8}{b}\right)^6 \cdot \frac{x^3}{x^6}$$

$$= {}^9C_6 \left(\frac{a}{2}\right)^3 \left(\frac{8}{b}\right)^6 \cdot [x^{-3}]$$

6. The term containing a^3b^4 in the expansion of $(a - 2b)^7$ is

(1) 3rd(2) 4th(3) 5th(4) 6th

Sol. Answer (3)

$$T_{r+1} = {}^7C_r (a)^{7-r} (-2b)^r$$

$$\text{Since, } a^{7-r} = a^3$$

$$\Rightarrow 7 - r = 3$$

$$r = 4$$

5th term contains a^3

7. The term independent of x in the expansion of $\left(x - \frac{3}{x^2}\right)^{18}$ is

(1) ${}^{18}C_6$ (2) ${}^{18}C_6 3^6$ (3) ${}^{18}C_{12}$ (4) ${}^{18}C_6 3^{12}$

Sol. Answer (2)

$$T_{r+1} = {}^{18}C_r (x)^{18-r} \cdot \left(\frac{-3}{x^2}\right)^r$$

$$= (-1)^r {}^{18}C_r 3^r x^{18-r-2r}$$

$$= (-1)^r \cdot {}^{18}C_r \cdot 3^r x^{18-3r}$$

...(1)

Since the term independent of x .

$$\text{If } 18 - 3r = 0$$

$$\Rightarrow r = 6$$

Put the value of r in (1), we obtain

$$\begin{aligned} T_7 &= (-1)^6 {}^{18}C_6 3^6 \\ &= {}^{18}C_6 3^6 \end{aligned}$$

8. In the expansion of $\left(\frac{x^3}{2} - \frac{2}{x^2}\right)^{12}$, 5th term from the end is

- (1) $-7920 x^{-4}$ (2) $7920 x^4$ (3) $7920 x^{-4}$ (4) $-7920 x^4$

Sol. Answer (3)

Fifth term from the end in the expansion of $\left(\frac{x^3}{2} - \frac{2}{x^2}\right)^{12}$ is equal to the 5th term from the beginning in the

expansion of $\left(\frac{-2}{x^2} + \frac{x^3}{2}\right)^{12}$

$$\begin{aligned} T_5 &= {}^{12}C_4 \left(\frac{-2}{x^2}\right)^{12-4} \cdot \left(\frac{x^3}{2}\right)^4 \\ &= \frac{{}^{12}C_4}{8} \frac{2^8}{x^{16}} \times \frac{x^{12}}{2^4} \\ &= \frac{12 \times 11 \times 10 \times 9}{4 \times 3 \times 2 \times 1} \times \frac{2^8}{2^4} \cdot x^{-4} \\ &= \frac{12 \times 11 \times 10 \times 9}{4 \times 3 \times 2 \times 1} \times 2^4 \times x^{-4} \end{aligned}$$

9. The coefficient of a^m and a^n (m, n are positive integer) in the expansion of $(1 + a)^{m+n}$ are

- (1) Unequal (2) Equal
(3) Reciprocal of each other (4) Additive inverse of each other

Sol. Answer (2)

$$\text{The coefficient of } a^m = {}^{m+n}C_m = \frac{{}^{m+n}C_n}{{}^{m+n}C_m}$$

$$\text{The coefficient of } a^n = {}^{m+n}C_n = \frac{{}^{m+n}C_m}{{}^{m+n}C_n}$$

Hence the coefficient of a^m = The coefficient of a^n

10. The number of terms in the expansion of $\{(a + 4b)^3 (a - 4b)^3\}^2$ is

- (1) 7 (2) 6 (3) 8 (4) 32

Sol. Answer (1)

$$\{(a+4b)^3(a-4b)^3\}^2$$

$$\{(a^2 - 16b^2)^3\}^2$$

$$= (a^2 - 16b^2)^6$$

which contains $6 + 1 = 7$ terms

11. If r^{th} term in the expansion of $\left(x^2 + \frac{1}{x}\right)^{12}$ is independent of x , then r is equal to

(1) 9

(2) 8

(3) 10

(4) 7

Sol. Answer (1)

$$T_r = {}^{12}C_{r-1} (x^2)^{12-(r-1)} \left(\frac{1}{x}\right)^{(r-1)} = {}^{12}C_{r-1} x^{26-2r-r+1}$$

$$= {}^{12}C_{r-1} x^{27-3r}$$

Since r^{th} term is void of x i.e., independent of x

$$\therefore 27 - 3r = 0$$

$$\Rightarrow r = 9.$$

12. The number of non-zero terms in the expansion of $(1+3\sqrt{2}a)^9 + (1-3\sqrt{2}a)^9$ is

(1) 10

(2) 5

(3) 9

(4) 6

Sol. Answer (2)

$$(1+3\sqrt{2}a)^9 + (1-3\sqrt{2}a)^9$$

$$= 2 \left\{ {}^9C_0 + {}^9C_2 (3\sqrt{2}a)^2 + {}^9C_4 (3\sqrt{2}a)^4 + {}^9C_6 (3\sqrt{2}a)^6 + {}^9C_8 (3\sqrt{2}a)^8 \right\}$$

Which contains only 5 terms

13. In the expansion of $\left(2 + \frac{1}{3x}\right)^n$, the coefficient of x^{-7} and x^{-8} are equal then n is equal to

(1) 51

(2) 52

(3) 55

(4) 56

Sol. Answer (3)

Coeff. of x^{-7} = coeff. of x^{-8}

$${}^nC_7 \cdot 2^{n-7} \cdot (3)^{-7}$$

$$= {}^nC_8 \cdot 2^{n-8} \cdot (3)^{-8}$$

$${}^nC_7 \times 6 = {}^nC_8$$

$$\frac{n!}{7!(n-7)!} \times 6 = \frac{n!}{8!(n-8)!}$$

$$n - 7 = 48$$

$$n = 55$$

14. If in the expansion of $(1 + px)^q$, $q \in N$, the coefficients of x and x^2 are 12 and 60 respectively, then p and q are

(1) 2, 6 (2) 6, 2 (3) -2, 6 (4) -6, 2

Sol. Answer (1)

In the expansion of $(1 + px)^q$, $T_2 = {}^qC_1 px$ and $T_3 = {}^qC_2 (px)^2$

Co-efficient of $x = pq = 12$

$$\text{Co-efficient of } x^2 = \frac{q(q-1)}{2} p = 60$$

$$\Rightarrow p^2 q^2 - p(pq) = 120$$

$$\Rightarrow 144 - 12p = 120$$

$$\Rightarrow 12p = 24$$

$$\Rightarrow p = 2, q = 6$$

15. The expansion of $\left(x^\alpha + \frac{1}{x^\beta}\right)^n$ has constant term, if

(1) $n\alpha$ is divisible by $n + \beta$

(2) $n\beta$ is divisible by $n + \alpha$

(3) $n\alpha$ is divisible by $\alpha + \beta$

(4) n is divisible by $\alpha + \beta$

Sol. Answer (3)

$$T_{r+1} = {}^nC_r (x^\alpha)^{n-r} (x^{-\beta})^r$$

${}^nC_r x^{\alpha n - r(\alpha + \beta)}$ is independent of x .

$$\text{If } \alpha n - r(\alpha + \beta) = 0$$

$$\Rightarrow \alpha n = r(\alpha + \beta)$$

$$\Rightarrow \alpha n \text{ is divisible by } (\alpha + \beta)$$

16. The number of rational terms in the expansion of $\left(\sqrt[3]{25} + \frac{1}{\sqrt[3]{25}}\right)^{20}$ is

(1) 2

(2) 7

(3) 6

(4) 19

Sol. Answer (2)

$$T_{r+1} = {}^{20}C_r (25)^{\frac{20-r}{3}} (25)^{\frac{r}{3}}$$

$$= {}^{20}C_r (25)^{\frac{20-2r}{3}}$$

Hence T_{r+1} to be rational,

$$r = 1, 4, 7, 10, 13, 16, 19$$

17. If in the expansion of $(1 + kx)^4$ the coefficient of x^3 is 32, then the value of k is equal to

(1) 2

(2) 4

(3) 8

(4) 1

Sol. Answer (1)

$$T_4 = T_{3+1} = {}^4C_3 k^3 x^3.$$

Then coefficient of $x^3 = 4k^3 = 32$

$$\Rightarrow k^3 = 8$$

$$\Rightarrow k = 2$$

18. If in the expansion of $\left(3 + \frac{x}{2}\right)^n$, the coefficient of x^7 and x^8 are equal, then the value of n is equal to

(1) 44

(2) 48

(3) 41

(4) 55

Sol. Answer (4)

$$T_8 = {}^nC_7 3^{n-7} \frac{x^7}{2^7}, T_9 = {}^nC_8 3^{n-8} \frac{x^8}{2^8}$$

$$\therefore {}^nC_7 \frac{3^{n-7}}{2^7} x^7 = {}^nC_8 \frac{3^{n-8}}{2^8} x^8$$

$$\Rightarrow 6 {}^nC_7 = {}^nC_8$$

$$\Rightarrow 6 \times 8 = n - 7$$

$$\Rightarrow n = 55$$

19. $\sqrt{5}\{(\sqrt{5}+1)^{50} - (\sqrt{5}-1)^{50}\}$ is

(1) An irrational number (2) 0

(3) A natural number

(4) A prime number

Sol. Answer (3)

Since $(x+a)^n - (x-a)^n = 2\{{}^nC_1 x^{n-1}a + {}^nC_3 x^{n-3}a^3 + \dots\}$ therefore

$$\begin{aligned} \left\{(\sqrt{5}+1)^{50} - (\sqrt{5}-1)^{50}\right\} &= 2\left\{{}^{50}C_1(\sqrt{5})^{49} + {}^{50}C_3(\sqrt{5})^{47} + \dots + {}^{50}C_{49}(\sqrt{5})\right\} \\ &= 2\sqrt{5}\left\{{}^{50}C_1 5^{24} + {}^{50}C_3 5^{23} + \dots + {}^{50}C_{49}\right\} \end{aligned}$$

Multiplying both sides by $\sqrt{5}$, we find that $\sqrt{5}\{(\sqrt{5}+1)^{50} - (\sqrt{5}-1)^{50}\}$ is a positive integer.

20. In the expansion of $(x+a)^5$, $T_2 : T_3 = 1 : 3$, then $x : a$ is equal to

(1) 1 : 2

(2) 2 : 1

(3) 2 : 3

(4) 3 : 2

Sol. Answer (3)

$$\frac{1}{3} = \frac{T_2}{T_3} = \frac{{}^5C_1 x^4 a}{{}^5C_2 x^3 a^2} = \frac{5x}{10a}$$

$$\Rightarrow \frac{x}{a} = \frac{10}{15}$$

$$\Rightarrow x : a = 2 : 3$$

21. If in the expansion of $\left(ax^2 + \frac{1}{bx}\right)^{11}$ the coefficient of x^7 is equal to the coefficient of x^{-7} in the expansion of

$\left(ax - \frac{1}{bx^2}\right)^{11}$, then ab is equal to

(1) 0

(2) 1

(3) -1

(4) 2

Sol. Answer (2)

$$\begin{aligned} T_{r+1} &= {}^{11}C_r (ax^2)^{11-r} \cdot \frac{1}{(bx)^r} \\ &= {}^{11}C_r \left(\frac{a^{11-r}}{b^r}\right) \cdot x^{22-2r-r} \end{aligned}$$

$$\Rightarrow 22 - 3r = 7$$

$$\Rightarrow r = 5$$

The coefficient of x^7 in

$$\left(ax^2 + \frac{1}{bx}\right)^{11}$$

$$= {}^{11}C_5 \frac{a^6}{b^5}$$

In the expansion of $\left(ax - \frac{1}{bx^2}\right)^{11}$

$$T_{r+1} = {}^{11}C_r (ax)^{11-r} \left[\frac{(-1)^r}{(bx^2)^r}\right]$$

$$= (-1)^r {}^{11}C_r \frac{a^{11-r}}{b^r} x^{11-3r}$$

$$\therefore 11 - 3r = -7$$

$$\Rightarrow r = 6$$

The coefficient of x^{-7} is ${}^{11}C_6 \frac{a^5}{b^6}$ and ${}^{11}C_5 \frac{a^6}{b^5} = {}^{11}C_6 \frac{a^5}{b^6}$

$$\Rightarrow ab = 1$$

22. The middle term in the expansion of $(1+x)^{2n}$ is

(1) ${}^{2n}C_n x^n$

(2) ${}^{2n}C_{n-1} x^{n+1}$

(3) ${}^{2n}C_{n+1} x^{n-1}$

(4) ${}^{2n}C_{n-1} x^n$

Sol. Answer (1)

$$\text{Middle term} = t_{n+1} = {}^{2n}C_n x^n$$

23. Coefficient of x^{12} in the expansion of $(1+x^2)^{50} \left(x + \frac{1}{x}\right)^{-10}$ is

(1) 41

(2) 40

(3) 43

(4) 44

Sol. Answer (2)

$$(1+x^2)^{50} \left(\frac{x^2+1}{x} \right)^{-10}$$

$$= x^{10} (1+x^2)^{40}$$

$\therefore$ The coefficient of x^{12} is ${}^{40}C_1 = 40$

24. The number of terms in the expansion of $(x^2 + 18x + 81)^{15}$ is

- (1) 15 (2) 16 (3) 30 (4) 31

Sol. Answer (4)

$$\text{Since } (x^2 + 18x + 81)^{15} = \{(x+9)^2\}^{15} = (x+9)^{30}$$

25. The term independent of x in the expansion of $\left(\sqrt[6]{x} - \frac{2}{\sqrt[3]{x}} \right)^{18}$ is

- (1) ${}^{18}C_8 2^{12}$ (2) ${}^{18}C_6 2^6$ (3) ${}^{18}C_6 2^8$ (4) ${}^{18}C_8 2^8$

Sol. Answer (2)

$$T_{r+1} = {}^{18}C_r x^{\frac{18-r}{6}} \cdot (-2)^r x^{-\frac{r}{3}}$$

$$= {}^{18}C_r (-2)^r x^{\frac{18-3r}{6}}$$

For independent of r

$$18 - 3r = 0$$

$$r = 6$$

26. The middle terms in the expansion of $(1+x)^{2n+1}$ is (are)

- (1) ${}^{2n+1}C_n x^n$ and ${}^{2n+1}C_{n+1} x^{n+1}$ (2) ${}^{2n+1}C_{n-1} x^{n+1}$ and ${}^{2n+1}C_{n+1} x^n$
 (3) ${}^{2n+1}C_n x^n$ only (4) ${}^{2n+1}C_{n+1} x^{n+1}$ only

Sol. Answer (1)

Number of terms $(2n+2)$

$\therefore$ Middle term is $(n+1)$ th and $n+2$

$$T_{n+1} = {}^{2n}C_n x^n$$

$$T_{n+2} = {}^{2n}C_{n+1} x^{n+1}$$

27. $(1.003)^4$ is nearly equal to

- (1) 1.012 (2) 1.0012 (3) 0.988 (4) 1.003

Sol. Answer (1)

$$(1.003)^4 = (1+0.003)^4 \approx 1 + 4 \times 0.003 = 1.012$$

28. The number of non-zero terms in the expansion of $(1+\sqrt{5})^6 + (\sqrt{5}-1)^6$ is

- (1) 3 (2) 4 (3) 5 (4) 0

Sol. Answer (2)

Since n is even

$$\therefore \text{The number of non-zero terms} = \frac{n}{2} + 1 = \frac{6}{2} + 1 = 4$$

29. The number of non-zero terms in the expansion of $(\sqrt{7} + 1)^{75} - (\sqrt{7} - 1)^{75}$ is

(1) 36

(2) 37

(3) 38

(4) 39

Sol. Answer (3)

Since n is odd

$\therefore$ The number of non-zero terms

$$= \frac{n+1}{2} = \frac{75+1}{2} = \frac{76}{2} = 38$$

30. The two consecutive term in the expansion of $(3 + 2x)^{74}$ which have equal coefficients, are

(1) 7th and 8th

(2) 11th and 12th

(3) 30th and 31st

(4) 31st and 32nd

Sol. Answer (3)

Suppose that coefficient of $T_r = {}^{74}C_{r-1}(3)^{75-r}(2x)^{r-1}$ and $T_{r+1} = {}^{74}C_r 3^{74-r}(2x)^r$ are equal.

$$\Rightarrow {}^{74}C_{r-1} 3^{75-r}(2)^{r-1} = {}^{74}C_r 3^{74-r}(2)^r$$

$$\Rightarrow r = 30$$

Hence, the coefficients of T_{30} and T_{31} are equal.

31. If the coefficient of r^{th} , $(r+1)^{\text{th}}$ and $(r+2)^{\text{th}}$ terms in the expansion of $(1+x)^{14}$ are in A.P., then the value of r is equal to

(1) 5 or 9

(2) 4 or 7

(3) 3 or 8

(4) 6 or 10

Sol. Answer (1)

If three consecutive coefficients in the expansion of $(1+x)^n$ are in A.P., then

$$r = \frac{n \pm \sqrt{n+2}}{2} = \frac{14 \pm \sqrt{14+2}}{2}$$

$$= 7 \pm 2$$

$$\Rightarrow 9 \text{ or } 5$$

32. Coefficient of $x^3y^{10}z^7$ in expansion of $(xy + yz)^{10}$ is

(1) ${}^{10}C_5$

(2) ${}^{10}C_4$

(3) ${}^{10}C_3$

(4) ${}^{10}C_{10}$

Sol. Answer (3)

$$(xy + yz)^{10} = y^{10}(x + z)^{10}$$

$\therefore$ Coeff. of $x^3z^7y^{10}$ is

$$= \frac{10!}{3!7!}$$

$$= {}^{10}C_3$$

33. The ratio of coeff. of x^3 and x^4 in expansion of $(1+x)^{12}$ is

- (1) $\frac{4}{9}$ (2) $\frac{1}{3}$ (3) $\frac{2}{3}$ (4) $\frac{1}{9}$

Sol. Answer (1)

$$(1+x)^{12}$$

$$\text{Coeff. of } x^3 = {}^{12}C_3 = \frac{12!}{3!5!}$$

$$\begin{aligned}\text{Coeff. of } x^4 &= {}^{12}C_4 = \frac{12!}{4!8!} \\ &= \frac{4}{9}\end{aligned}$$

34. Given positive integers $r > 1$, $n > 2$ and the coefficients of $(3r)^{\text{th}}$ and $(r+2)^{\text{th}}$ term in the expansion of $(1+x)^{2n}$ are equal, then, which of the following relation is correct?

- (1) $n = 2r$ (2) $n = 3r$ (3) $n = 2r + 1$ (4) $n = r$

Sol. Answer (1)

$$T_{3r} = {}^{2n}C_{3r-1}x^{3r-1} \text{ and } T_{r+2} = {}^{2n}C_{r+1}x^{r+1}$$

$$\text{Given } {}^{2n}C_{3r-1} = {}^{2n}C_{r+1}$$

$$3r - 1 = r + 1 \quad \text{or} \quad 3r - 1 = 2n - r - 1$$

$$\Rightarrow 2r = 2 \quad \text{or} \quad 4r = 2n$$

$$\Rightarrow r = 1 \quad \text{or} \quad n = 2r$$

$$\text{As } r > 1 \quad \therefore n = 2r$$

35. The coefficient of x^5 in the expansion of $(1+x^2)^5(1+x)^4$ is

- (1) 61 (2) 59 (3) 0 (4) 60

Sol. Answer (4)

$$(1+x^2)^5(1+x)^4$$

$$\text{Coeff. of } x^5 = {}^5C_1 \cdot {}^4C_3 + {}^5C_2 \cdot {}^4C_1$$

$$= 20 + 40 = 60$$

36. If $(r+1)^{\text{th}}$ term in the expansion of $\left(\frac{a^3}{3} - \frac{2}{a^2}\right)^{10}$ contains a^{20} then the value of r is equal to

- (1) 3 (2) 2 (3) 4 (4) 1

Sol. Answer (2)

$$T_{r+1} = {}^{10}C_r \left(\frac{a^3}{3}\right)^{10-r} \left(-\frac{2}{a^2}\right)^r$$

$$= {}^{10}C_r \frac{a^{30-3r-2r}(-2)^r}{3^{10-r}} \text{ contains } a^{20} \text{ if } 30 - 3r - 2r = 20$$

$$\Rightarrow r = 2$$

37. If in the expansion of $(1+x)^n$, fifth term is 4 times the fourth term and fourth term is 6 times the third term, then the value of n and x is

(1) 11, 2

(2) 2, 11

(3) 3, 12

(4) 12, 3

Sol. Answer (1)

$${}^nC_4 x^4 = 4 \times {}^nC_3 x^3 \quad \dots(i)$$

$${}^nC_3 x^3 = 6 \times {}^nC_2 x^2 \quad \dots(ii)$$

On solving $n = 11, x = 2$

38. Coefficient of $x^6 y^3$ in the expansion of $(x+y)^9$ is

(1) 36

(2) 16

(3) 84

(4) 100

Sol. Answer (3)

$$(x+y)^9 \text{ coefficient of } x^6 y^3 = \frac{9!}{6!3!}$$

39. The number of terms in the expansion of $(4x^2 + 9y^2 + 12xy)^6$ is

(1) 2

(2) 12

(3) 13

(4) 28

Sol. Answer (3)

$(2x+3y)^{12}$ so 13 terms

40. The middle term in the expansion of $\left(2x - \frac{1}{3x}\right)^{10}$ is

(1) $-{}^{10}C_5 \left(\frac{2}{3}\right)^5$

(2) ${}^{10}C_5 \left(\frac{2}{3}\right)^5$

(3) ${}^{10}C_6 2^4 3^{-6}$

(4) $-{}^{10}C_6 2^4 \cdot 3^{-6}$

Sol. Answer (1)

$$\text{Middle term} = t_6 = {}^{10}C_5 2^5 \left(-\frac{1}{3}\right)^5$$

41. The term independent of x in the expansion of $\left(ax + \frac{b}{x}\right)^{14}$ is

(1) ${}^{14}C_5 a^9 b^5$

(2) ${}^{14}C_6 a^8 b^6$

(3) ${}^{14}C_7 a^7 b^7$

(4) ${}^{14}C_8 a^6 b^8$

Sol. Answer (3)

$$\left(ax + \frac{b}{x}\right)^{14}$$

the term independent of $x = {}^{14}C_7 a^7 b^7$

42. Middle term in the expansion of $\left(x - \frac{1}{2x}\right)^{12}$ is

(1) $\frac{1}{5}$

(2) $\frac{2}{5}$

(3) $\frac{231}{16}$

(4) $\frac{1}{16}$

Sol. Answer (3)

T_7 = middle term

$$= {}^{12}C_6 (x)^{12-6} \left(-\frac{1}{2x}\right)^6 = \frac{231}{16}$$

43. In the binomial expansion of $(a - b)^n$, $n \geq 5$, the sum of the 5th and 6th term is zero, then $\frac{a}{b}$ equals

(1) $\frac{n-5}{6}$

(2) $\frac{n-4}{5}$

(3) $\frac{5}{n-4}$

(4) $\frac{6}{n-5}$

Sol. Answer (2)

$$T_5 + T_6 = {}^nC_4 a^{n-4} (-b)^4 + {}^nC_5 a^{n-5} (-b)^5 = 0$$

$$\Rightarrow \frac{a^{n-4} b^4}{a^{n-5} b^5} = \frac{{}^nC_5}{{}^nC_4} \Rightarrow \frac{a}{b} = \frac{n-5+1}{5} = \frac{n-4}{5}$$

44. The number of term in $(x + y)^{100} + (x - y)^{100}$ is equal to

(1) 51

(2) 101

(3) 202

(4) 50

Sol. Answer (1)

$$(x + y)^{100} = {}^{100}C_0 x^{100} + {}^{100}C_1 x^{99} y + \dots + {}^{100}C_{100} y^{100}$$

$$(x - y)^{100} = {}^{100}C_0 x^{100} + {}^{100}C_1 x^{99} (-y) + \dots + {}^{100}C_{100} y^{100}$$

$$(x + y)^{100} + (x - y)^{100} = 2[{}^{100}C_0 x^{100} + {}^{100}C_2 x^{98} y^2 + \dots + {}^{100}C_{100} y^{100}]$$

Hence total terms = 51.

45. The sum of the binomial coefficients of $\left(2x + \frac{1}{x}\right)^n$ is equal to 256. Then the constant term in the expansion is

(1) 1120

(2) 2110

(3) 1210

(4) 2210

Sol. Answer (1)

Sum of the binomial coefficients = $2^n = 256$

$$\Rightarrow n = 8$$

Let T_{r+1} is the constant term then

$$T_{r+1} = {}^nC_r (2x)^{n-r} \left(\frac{1}{x}\right)^r$$

$$T_{r+1} = {}^nC_r 2^{n-r} x^{n-r} x^{-r} = {}^nC_r 2^{n-r} x^{n-2r}$$

$$\text{For constant term } n - 2r = 0 \Rightarrow r = \frac{n}{2} = \frac{8}{2} = 4$$

$$\text{Hence the constant term} = {}^8C_4 \cdot 2^{8-4} = {}^8C_4 \times 16 = 1120$$

46. If the coefficients of 2nd, 3rd and 4th terms in the expansion of $(1+x)^{2n}$ are in A.P., then

$$(1) \ 2n^2 - 9n + 7 = 0 \quad (2) \ 2n^2 + 5n + 7 = 0 \quad (3) \ n^2 - 9n + 7 = 0 \quad (4) \ n^2 + 9n - 7 = 0$$

Sol. Answer (1)

$$(1+x)^{2n} = 1 + {}^{2n}C_1 x + {}^{2n}C_2 x^2 + {}^{2n}C_3 x^3 + \dots + {}^{2n}C_{2n} x^{2n}$$

Given that

$${}^{2n}C_1, {}^{2n}C_2, {}^{2n}C_3 \text{ are in A.P.}$$

$$\Rightarrow 2 \cdot {}^{2n}C_2 = {}^{2n}C_1 + {}^{2n}C_3$$

$$\Rightarrow 2 \cdot \frac{2n(2n-1)}{2} = 2n + \frac{2n(2n-1)(2n-2)}{3 \times 2}$$

$$\Rightarrow (2n-1) = 1 + \frac{(2n-1)(2n-2)}{3 \times 2}$$

$$\Rightarrow 2n-2 = \frac{(2n-1)(n-1)}{3}$$

$$\Rightarrow 6n-6 = 2n^2-3n+1$$

$$\Rightarrow 2n^2-9n+7=0$$

47. $\sum_{r=0}^{n-1} \frac{{}^nC_r}{{}^nC_r + {}^nC_{r+1}}$ equals

$$(1) \ \frac{n}{2}$$

$$(2) \ \frac{n+1}{2}$$

$$(3) \ \frac{n(n+1)}{2}$$

$$(4) \ \frac{n(n-1)}{2(n+1)}$$

Sol. Answer (1)

$$\sum_{r=0}^{n-1} \frac{{}^nC_r}{{}^nC_r + {}^nC_{r+1}} = \sum_{r=0}^{n-1} \frac{1}{1 + \frac{{}^nC_{r+1}}{{}^nC_r}} = \sum_{r=0}^{n-1} \frac{1}{1 + \frac{\frac{n!}{(r+1)!(n-r)!}}{\frac{n!}{r!(n-r)!}}}$$

$$= \sum_{r=0}^{n-1} \frac{1}{1 + \frac{n-r}{r+1}} = \sum_{r=0}^{n-1} \frac{r+1}{n+1}$$

$$= \frac{1}{n+1} [1+2+3+\dots+n] = \frac{1}{n+1} \cdot \frac{(n+1)(n)}{2} = \frac{n}{2}$$

48. The greatest term in the expansion of $(3 - 5x)^{15}$, when $x = \frac{1}{5}$, is

- (1) 6th (2) 5th (3) 5th & 6th (4) 4th & 5th

Sol. Answer (4)

Let T_{r+1} is the greatest term

$$\Rightarrow |T_{r+1}| \geq |T_r|$$

$$\left| \frac{T_{r+1}}{T_r} \right| \geq 1$$

$$\Rightarrow \left| \frac{{}^{15}C_r 3^{15-r} (-5x)^r}{{}^{15}C_{r-1} 3^{16-r} (-5x)^{r-1}} \right| \geq 1$$

$$\Rightarrow \left(\frac{15-r+1}{r} \right) \left| \frac{-5x}{3} \right| \geq 1 \Rightarrow \left(\frac{16-r}{r} \right) \left| \frac{5x}{3} \right| \geq 1$$

$$\Rightarrow \left(\frac{16-r}{r} \right) \left| \frac{5 \times \frac{1}{5}}{3} \right| \geq 1 \Rightarrow \left(\frac{16-r}{r} \right) \left(\frac{1}{3} \right) \geq 1$$

$$16 - r \geq 3r \Rightarrow 16 \geq 4r$$

$$r \leq 4$$

Hence $r = 4$

Greatest term = $T_r, T_{r+1} = T_4, T_5$

Hence 4th and 5th term are greatest.

49. In the expansion of $(y^{\frac{1}{5}} + x^{\frac{1}{10}})^{55}$, the number of terms free of radical sign are

- (1) 5 (2) 6 (3) 50 (4) 56

Sol. Answer (2)

$$T_{r+1} = {}^{55}C_r (y^{\frac{1}{5}})^{55-r} (x^{\frac{1}{10}})^r$$

$${}^{55}C_r y^{\frac{55-r}{5}} x^{\frac{r}{10}}$$

The term is free from radical sign if

$$\frac{55-r}{5} \text{ and } \frac{r}{10} \text{ both are integers}$$

$\Rightarrow r = 0, 15, 25, 35, 45, 55$ makes both integers hence total 6 terms will be free from radical sign.

50. The term independent of x in $(1+x+2x^3)\left(\frac{3x^2}{2} - \frac{1}{3x}\right)^9$ is

- (1) 25/54 (2) 17/54 (3) 1/6 (4) -17/54

Sol. Answer (2)

The term independent of x in $(1 + x + 2x^3) \left(\frac{3x^2}{2} - \frac{1}{3x} \right)^9$

$$= \text{Coefficient of } x^0 \text{ in } \left(\frac{3x^2}{2} - \frac{1}{3x} \right)^9 + \text{Coefficient of } x^{-1} \text{ in } \left(\frac{3x^2}{2} - \frac{1}{3x} \right)^9 + 2 \times \text{coefficient of } x^{-3} \text{ in } \left(\frac{3x^2}{2} - \frac{1}{3x} \right)^9$$

To find these coefficients let us find general term in $\left(\frac{3x^2}{2} - \frac{1}{3x} \right)^9$, that is given by

$$T_{r+1} = {}^9C_r \left(\frac{3x^2}{2} \right)^{9-r} \left(-\frac{1}{3x} \right)^r = {}^9C_r (-1)^r \left(\frac{3}{2} \right)^{9-r} \left(\frac{1}{3} \right)^r x^{18-2r-r}$$

$$= {}^9C_r (-1)^r \left(\frac{3}{2} \right)^{9-r} \left(\frac{1}{3} \right)^r x^{18-3r}$$

For constant term $18 - 3r = 0 \Rightarrow r = 6$

For coefficient of x^{-1} , $18 - 3r = -1 \Rightarrow 19 = 3r$

$\Rightarrow r = \frac{19}{3}$ which is not an integer, hence coefficient in this case is zero.

For coefficient of x^{-3} , $18 - 3r = -3$

$$3r = 21$$

$$r = 7$$

$\therefore$ The term independent of x

$$= {}^9C_6 \left(\frac{3}{2} \right)^3 \cdot \left(-\frac{1}{3} \right)^6 + 2 \cdot {}^9C_7 \left(\frac{3}{2} \right)^2 \cdot \left(-\frac{1}{3} \right)^7$$

$$= \frac{17}{54}$$

51. Find the coefficient of x^3 in the expansion of $(1 + x + 2x^2) \left(2x^2 - \frac{1}{3x} \right)^9$.

(1) $-\frac{224}{9}$

(2) $-\frac{112}{27}$

(3) $-\frac{224}{27}$

(4) $-\frac{112}{27}$

Sol. Answer (3)

The expression is

$$(1 + x + 2x^2) \left(2x^2 - \frac{1}{3x} \right)^9$$

$$= \left(2x^2 - \frac{1}{3x} \right)^9 + x \left(2x^2 - \frac{1}{3x} \right)^9 + 2x^2 \left(2x^2 - \frac{1}{3x} \right)^9$$

In the expansion of

$$\left(2x^2 - \frac{1}{3x}\right)^9$$

General term

$$T_{r+1} = {}^9C_r \cdot (2x^2)^{9-r} \cdot \left(-\frac{1}{3x}\right)^r$$

$$= {}^9C_r \cdot 2^{18-2r} \cdot x^{9-r} \cdot \left(-\frac{1}{3}\right)^r \cdot x^{-r}$$

$$= {}^9C_r \cdot 2^{9-r} \cdot \left(-\frac{1}{3}\right)^r \cdot x^{18-3r}$$

$\therefore$ we are looking for the coefficient of x^3

$$\therefore 18 - 3r = 3$$

$$\Rightarrow 3r = 15$$

$$\Rightarrow r = 5$$

For the coefficient of x^2 ,

$$18 - 3r = 2$$

$$\Rightarrow r = \frac{16}{3} \text{ (Not possible)}$$

For the coefficient of x

$$18 - 3r = 1$$

$$\Rightarrow r = \frac{17}{3} \text{ (Not possible)}$$

Now, the coefficient of x^3 in

$$\left(2x^2 - \frac{1}{3x}\right)^9 + x\left(2x^2 - \frac{1}{3x}\right)^9 + 2x^2\left(2x^2 - \frac{1}{3x}\right)^9$$

$$= \text{Coefficient of } x^3 \text{ in } \left(2x^2 - \frac{1}{3x}\right)^9 +$$

$$\text{coefficient of } x^2 \left(2x^2 - \frac{1}{3x}\right)^9 \text{ in } + 2 \times \text{Coefficient of } x \text{ in } \left(2x^2 - \frac{1}{3x}\right)^9$$

$$= {}^9C_5 \cdot 2^{9-5} \cdot \left(-\frac{1}{3}\right)^5 + 0 + 2(0)$$

$$= \frac{9 \times 8 \times 7 \times 6}{4 \times 3 \times 2} \cdot 2^4 \cdot -\frac{1}{3^5} = -\frac{224}{27}$$

52. If in the expansion of $(1+x)^m(1-x)^n$, the coefficient of x and x^2 are 3 and -6 respectively, then the value of n is

(1) 6

(2) 9

(3) 12

(4) 24

Sol. Answer (2)

$$(1+x)^m(1-x)^n$$

$$= ({}^mC_0 + {}^mC_1x + {}^mC_2x^2 + \dots + {}^mC_mx^m)({}^nC_0 - {}^nC_1x + {}^nC_2x^2 + \dots + (-1)^n {}^nC_n)$$

$$\text{Coefficient of } x^2 = {}^mC_0 {}^nC_2 - {}^mC_1 {}^nC_1 + {}^mC_2 {}^nC_0 = -6 \quad \dots(i)$$

$$\text{Coefficient of } x = {}^mC_0 (-{}^nC_1) + {}^mC_1 {}^nC_0 = 3 \quad \dots(ii)$$

$$\text{By (i)} \quad \frac{n(n-1)}{2} - mn + \frac{m(m-1)}{2} = -6 \quad \dots(iii)$$

$$\text{By (ii)} \quad -n + m = 3 \quad \dots(iv)$$

By (iii) and (iv)

$$\frac{n(n-1)}{2} - (n+3)n + \frac{(n+3)(n+2)}{2} = -6$$

$$n(n-1) - 2n(n+3) + (n+3)(n+2) = -12$$

$$n^2 - n - 2n^2 - 6n + n^2 + 5n + 6 = -12$$

$$-2n = -18$$

$$\Rightarrow n = 9$$

53. If 6th term of $\left[2^{\log_2 \sqrt{9^{x-1}+7}} + \frac{1}{2^{\frac{1}{5} \log_2 (3^{x-1}+1)}} \right]^7$ is 84, then $x =$

(1) 4 or 3

(2) 3 or 1

(3) 2 or 1

(4) 1

Sol. Answer (3)

$$T_6 = T_{5+1} = {}^7C_5 \left(2^{\log_2 \sqrt{9^{x-1}+7}} \right)^2 \left(\frac{1}{2^{\frac{1}{5} \log_2 (3^{x-1}+1)}} \right)^5 = 84$$

$${}^7C_5 (9^{x-1} + 7) \cdot \left(\frac{1}{3^{x-1} + 1} \right) = 84$$

$$\frac{7 \times 6}{2} (9^{x-1} + 7) \cdot \left(\frac{1}{3^{x-1} + 1} \right) = 84$$

$$\frac{9^{x-1} + 7}{3^{x-1} + 1} = 4 \Rightarrow 9^{x-1} + 7 = 4(3^{x-1} + 1)$$

$$\Rightarrow \frac{9^x}{9} + 7 = 4(3^{x-1} + 1)$$

$$9^x + 63 = 36 \left(\frac{3^x}{3} + 1 \right)$$

$$\Rightarrow 9^x + 63 = 12 \cdot 3^x + 36$$

$$\text{Let } 3^x = y$$

$$y^2 = 63 = 12y + 36$$

$$y^2 - 12y + 27 = 0$$

$$\Rightarrow y = 3, y = 9$$

$$\Rightarrow 3^x = 3 \text{ or } 9 \Rightarrow x = 1, 2$$

[Properties of Binomial Coefficients, Differentiation and Integration, Multinomial Theorem]

54. The number of terms which are not similar in the expansion of $(l + m + n)^6$ is

(1) 7

(2) 42

(3) 28

(4) 21

Sol. Answer (3)

$$(l + m + n)^6 = \{l + (m + n)\}^6$$

$$= {}^6C_0 l^6 + {}^6C_1 l^5 (m + n) + {}^6C_2 l^4 (m + n)^2 + {}^6C_3 l^3 (m + n)^3 + {}^6C_4 l^2 (m + n)^4 + {}^6C_5 l (m + n)^5 + {}^6C_6 (m + n)^6 \dots (1)$$

Since the expansion $(m + n)^k$ contains $(k + 1)$ terms therefore

R.H.S. of (1) there are

$$1 + 2 + 3 + 4 + 5 + 6 + 7 = 28 \text{ terms}$$

55. The number of terms in the expansion of $(a + b + c)^{12}$ is

(1) 90

(2) 91

(3) 81

(4) 80

Sol. Answer (2)

$$\text{The number of terms in the expansion of } (a + b + c)^{12} = {}^{12+3-1}C_{3-1} = {}^{14}C_2 = 7 \times 13 = 91$$

$$\text{or } 1 + 2 + 3 + \dots + 13 = \frac{13 \times 14}{2} = 13 \times 7 = 91$$

56. The value of ${}^{13}C_7 + {}^{13}C_8 + {}^{13}C_9 + {}^{13}C_{10} + {}^{13}C_{11} + {}^{13}C_{12} + {}^{13}C_{13}$ is equal to

(1) 2^{12}

(2) 2^{11}

(3) $\frac{2^{13} + 1}{2}$

(4) $\frac{2^{13} - 1}{2}$

Sol. Answer (1)

$${}^{13}C_7 + {}^{13}C_8 + \dots + {}^{13}C_{13} = ?$$

$${}^{13}C_0 + {}^{13}C_1 + \dots + {}^{13}C_6 + {}^{13}C_7 + {}^{13}C_8 + \dots + {}^{13}C_{13} = 2^{13}$$

$$\therefore {}^{13}C_7 + {}^{13}C_8 + \dots + {}^{13}C_{13} = 2^{12}$$

57. If sum of the coefficients in the expansion of $(2x + 3y - 2z)^n$ is 2187 then the greatest coefficient in the expansion of $(1 + x)^n$

(1) 30

(2) 40

(3) 28

(4) 35

Sol. Answer (4)

Sum is obtained by putting $x = y = z = 1$

$$\text{Sum} = (2 + 3 - 2)^n$$

$$\Rightarrow 3^n = 2187 \Rightarrow n = 7$$

$$\text{Greatest coefficient} = {}^7C_3 = {}^7C_4 = \frac{7 \times 6 \times 5}{3 \times 2 \times 1} = 35$$

58. If $(1+x)^n = \sum_{r=0}^n C_r \cdot x^r$, then $\left(1 + \frac{C_1}{C_0}\right) \left(1 + \frac{C_2}{C_1}\right) \dots \left(1 + \frac{C_n}{C_{n-1}}\right) =$

- (1) $\frac{n^{n-1}}{(n+1)!}$ (2) $\frac{(n+1)^{n+1}}{(n+1)!}$ (3) $\frac{(n+1)^n}{n!}$ (4) $\frac{(n+1)^{n+1}}{n!}$

Sol. Answer (3)

The general term $= \left(1 + \frac{C_r}{C_{r-1}}\right) = T_r$

The given product can be written as $T_1 T_2 T_3 \dots T_n$

$$T_r = 1 + \frac{C_r}{C_{r-1}} = 1 + \frac{{}^nC_r}{{}^nC_{r-1}} = 1 + \frac{n-r+1}{r} = \frac{n+1}{r}$$

$$\text{Product} = \frac{(n+1)}{1} \cdot \frac{(n+1)}{2} \dots \frac{(n+1)}{n} = \frac{(n+1)^n}{n!}$$

59. The sum of the last eight coefficients in the expansion of $(1+x)^{16}$ is equal to

- (1) 2^{15} (2) 2^{14} (3) $2^{15} - \frac{1}{2} \frac{16!}{(8!)^2}$ (4) 2^{16}

Sol. Answer (3)

$$(1+x)^{16} = C_0 + C_1x + C_2x^2 + \dots + C_{16}x^{16}, \text{ where } C_r = {}^{16}C_r$$

$$\text{Let } S = C_{16} + C_{15} + \dots + C_9$$

$$\text{Again } 2S = 2C_{16} + 2C_{15} + \dots + 2C_9$$

$$= (C_{16} + C_{16}) + (C_{15} + C_{15}) + \dots + (C_9 + C_9)$$

$$= (C_0 + C_{16}) + (C_1 + C_{15}) + \dots + (C_7 + C_9)$$

$$= (C_0 + C_1 + \dots + C_7 + C_8 + \dots + C_{16}) - C_8$$

$$= 2^{16} - C_8 = 2^{16} - \frac{16!}{8!8!}$$

$$S = 2^{15} - \frac{1}{2} \frac{16!}{8!8!}$$

60. The coefficient of $x^3y^4z^5$ in the expansion of $(xy + yz + zx)^6$ is

- (1) 60 (2) 120 (3) 6! (4) 0

Sol. Answer (1)

$$x^3y^4z^5 = (xy)^a(yz)^b(zx)^c = x^{a+c}y^{b+a}z^{c+b}$$

$\Rightarrow$ Comparing we get

$$a + c = 3 \quad \dots(i)$$

$$b + a = 4 \quad \dots(ii)$$

$$c + b = 5 \quad \dots(iii)$$

Adding (i), (ii), (iii), we get $a + b + c = 6$

$$\Rightarrow b = 3, c = 2, a = 1$$

$$\text{Coefficient} = \frac{6!}{3!2!1!} = \frac{6 \times 5 \times 4 \times 3 \times 2}{3 \times 2 \times 2} = 60$$

61. In the expansion of $(3x + 2y - z)^8$, the coefficients of $x^2y^3z^3$ is

(1) 10084

(2) -40320

(3) 20160

(4) -43280

Sol. Answer (2)

$$(3x + 2y - z)^8 = \sum_{a+b+c=8} \frac{8!}{a!b!c!} (3x)^a (2y)^b (-z)^c$$

$\Rightarrow$ The coefficient of $x^a y^b z^c$

$$= \frac{8!}{a!b!c!} 3^a 2^b (-1)^c$$

Here $a = 2, b = 3, c = 3$

$$\begin{aligned} \text{Coefficient} &= \frac{8!}{2!3!3!} \times 3^2 \times 2^3 \times (-1)^3 \\ &= \frac{-8!}{2 \times 3 \times 2 \times 3 \times 2} \times 9 \times 8 = -40320 \end{aligned}$$

62. If n is an integer greater than 1, then $a - {}^nC_1(a-1) + {}^nC_2(a-2) - \dots + (-1)^n(a-n) =$

(1) a

(2) 0

(3) a^2

(4) 2^n

Sol. Answer (2)

The given expression can be written as

$$\begin{aligned} &a(1 - {}^nC_1 + {}^nC_2 - {}^nC_3 + \dots + (-1)^n {}^nC_n) + ({}^nC_1 - 2 \cdot {}^nC_2 + 3 \cdot {}^nC_3 - 4 \cdot {}^nC_4 + \dots) \\ &= a \times 0 + 0 = 0 \end{aligned}$$

63. $\frac{C_0}{1} + \frac{C_1}{2} + \frac{C_2}{3} + \dots + \frac{C_{100}}{101}$ equals

(1) $\frac{2^{101}}{101}$

(2) $\frac{2^{101}-1}{101}$

(3) $\frac{3^{101}}{101}$

(4) $\frac{3^{101}-1}{101}$

Sol. Answer (2)

As we know that

$$\frac{(1+x)^{n+1}-1}{n+1} = C_0x + \frac{C_1x^2}{2} + \frac{C_2x^3}{3} + \dots + \frac{C_nx^{n+1}}{n+1}$$

At $x = 1$

$$\text{We find that } C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \dots + \frac{C_n}{n+1} = \frac{2^{n+1}-1}{n+1}$$

For $n = 100$

$$\text{We get } C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \dots + \frac{C_{100}}{101} = \frac{2^{101}-1}{101}$$

64. If $k = \sum_{r=0}^n \frac{1}{{}^nC_r}$, then $\sum_{r=0}^n \frac{r}{{}^nC_r}$ is equal to

(1) nk (2) $\frac{nk}{2}$ (3) $(n-1)k$ (4) $\frac{nk}{3}$ **Sol.** Answer (2)

$$k = \sum_{r=0}^n \frac{1}{{}^nC_r} \quad \dots (i)$$

$$\sum_{r=0}^n \frac{r}{{}^nC_r} = \sum_{r=0}^n \frac{r-n+n}{{}^nC_r} = \sum_{r=0}^n \frac{n}{{}^nC_r} - \sum_{r=0}^n \frac{n-r}{{}^nC_r}$$

$$\Rightarrow \sum_{r=0}^n \frac{r}{{}^nC_r} = nk - \sum_{r=0}^n \frac{(n-r)}{{}^nC_{n-r}} \text{ by (i)}$$

$$= nk - \sum_{r=0}^n \frac{r}{{}^nC_r}$$

$$\Rightarrow 2 \sum_{r=0}^n \frac{r}{{}^nC_r} = nk \Rightarrow \sum_{r=0}^n \frac{r}{{}^nC_r} = \frac{nk}{2}$$

65. If $x + y = 1$, then $\sum_{r=0}^n r^2 {}^nC_r x^r y^{n-r}$ equals

(1) nxy (2) $nx(x + yn)$ (3) $nx(nx + y)$

(4) 1

Sol. Answer (3)

$$S = \sum_{r=0}^n r^2 x^r y^{n-r}$$

$$\text{As } (y+x)^n = {}^nC_0 y^n x^0 + {}^nC_1 y^{n-1} x^1 + {}^nC_2 y^{n-2} x^2 + \dots + {}^nC_n x^n$$

Differentiating w.r.t. (x) keeping y as constant

$$n(y+x)^{n-1} = {}^nC_1 y^{n-1} + 2 {}^nC_2 y^{n-2} x + \dots + n {}^nC_n x^{n-1}$$

$$nx(y+x)^{n-1} = {}^nC_1 y^{n-1} x + 2 {}^nC_2 y^{n-2} x^2 + \dots + n {}^nC_n x^n$$

Again differentiating w.r.t. (x) keeping y constant

$$\begin{aligned} n[x.(n-1)(y+x)^{n-2} + (y+x)^{n-1}] \\ = {}^nC_1 y^{n-1} + 2^2 {}^nC_2 y^{n-2} x + 3^2 {}^nC_3 y^{n-3} x^2 + \dots + n^2 {}^nC_n x^{n-1} \end{aligned}$$

But $x + y = 1$

$$\Rightarrow n[x(n-1) + 1] = S$$

$$\Rightarrow n[nx + 1 - x] = S$$

$$\Rightarrow S = n[nx + y]$$

66. $\sum_{r=1}^n r ({}^nC_r - {}^nC_{r-1})$ is equal to

(1) $2^n + n + 1$

(2) $2^n - n + 1$

(3) $n - 2^n + 1$

(4) $n - 2^n - 1$

Sol. Answer (3)

$$\begin{aligned} S &= \sum_{r=1}^n r ({}^nC_r - {}^nC_{r-1}) = \sum_{r=1}^n r {}^nC_r - \sum_{r=1}^n r {}^nC_{r-1} = n \cdot 2^{n-1} - \sum_{r=1}^n r {}^nC_{r-1} \\ S &= n \cdot 2^{n-1} - [{}^nC_0 + 2 {}^nC_1 + 3 {}^nC_2 + 4 {}^nC_3 + \dots + n {}^nC_{n-1}] \end{aligned} \quad \dots(i)$$

Let $(1+x)^n = C_0 + C_1 x + C_2 x^2 + \dots + C_n x^n$ where $C_r = {}^nC_r$

$$\Rightarrow x(1+x)^n = C_0 x + C_1 x^2 + C_2 x^3 + \dots + C_n x^{n+1}$$

Differentiating

$$(1+x)^n + n(1+x)^{n-1} = C_0 + 2C_1 x + 3C_2 x^2 + \dots + (n+1)C_n x^n$$

Putting $x = 1$

$$2^n + n \cdot 2^{n-1} = C_0 + 2C_1 + 3C_2 + \dots + (n+1)C_n$$

$$\Rightarrow 2^n + n \cdot 2^{n-1} = C_0 + 2C_1 + 3C_2 + \dots + nC_{n-1} + (n+1)C_n$$

$$\Rightarrow (C_0 + 2C_1 + 3C_2 + \dots + nC_{n-1}) = 2^n + n \cdot 2^{n-1} - (n+1)C_n \dots \dots \dots \quad \dots(ii)$$

If $C_r = {}^nC_r$

Then we may write that

$$\begin{aligned} & {}^nC_0 + 2 {}^nC_1 + 3 {}^nC_2 + \dots + n {}^nC_{n-1} \\ &= 2^n + n \cdot 2^{n-1} - (n+1)C_n \end{aligned} \quad \dots(iii)$$

By (i) and (iii)

$$S = n \cdot 2^{n-1} - [2^n + n \cdot 2^{n-1} - n - 1]$$

$$S = n + 1 - 2^n$$

67. The coefficient of x^4 in the expansion of $(1 + x + x^2 + x^3)^{11}$ is

(1) 900

(2) 909

(3) 990

(4) 999

Sol. Answer (3)

$$S = (1 + x + x^2 + x^3)^{11} = \left(\frac{1 - x^4}{1 - x} \right)^{11} = (1 - x^4)^{11} (1 - x)^{-11}$$

$$= (1 + {}^{11}C_1(-x)^4 + \dots) \left(1 + 11x + \frac{11 \times 12}{2!} x^2 + \dots \right)$$

$$\text{Coefficient of } x^4 = \frac{11 \times 12 \times 13 \times 14}{4!} - 11 = 990$$

[Miscellaneous]

68. The number of zeroes at the end of $(101)^{11} - 1$ is

(1) 8

(2) 4

(3) 6

(4) 2

Sol. Answer (4)

$$(101)^{11} - 1 = (1 + 100)^{11} - 1$$

$$= 1 + {}^{11}C_1(100) + {}^{11}C_2(100)^2 + \dots + {}^{11}C_{11}(100)^{11} - 1$$

$$= 100[{}^{11}C_1 + {}^{11}C_2 100 + {}^{11}C_3 (100)^2 + \dots + {}^{11}C_{11} (100)^{10}]$$

$$= 100 (\text{odd number})$$

$\therefore$ The number of zeros of the end is 2.

69. For all natural number n , $2^{2n} \cdot 3^{2n} - 1 - 35n$ is divisible by

(1) $(35)^3$ (2) $(35)^2$ (3) $(35)^4$ (4) $(35)^5$

Sol. Answer (2)

$$2^{2n} \cdot 3^{2n} - 1 - 35n$$

$$(6)^{2n} - 1 - 35n$$

$$= (36)^n - 35n - 1$$

$$= (1 + 35)^n - 35n - 1$$

$$= {}^nC_0 + {}^nC_1 35 + {}^nC_2 35^2 + \dots + {}^nC_n 35^n - 35n - 1$$

$$= 35^2 ({}^nC_2 + {}^nC_3 35 + \dots + {}^nC_n 35^{n-2})$$

70. If $3^{2n+2} - 8n - 9$ is divided by 64, then remainder is

(1) 1

(2) 0

(3) 7

(4) 2

Sol. Answer (2)

$$(3)^{2(n+1)} - 8n - 9 = (8 + 1)^{n+1} - 8n - 9$$

$$= {}^{n+1}C_0 8^{n+1} + {}^{n+1}C_1 8^n + \dots + {}^{n+1}C_n 8 + {}^{n+1}C_{n+1} - 8n - 9$$

$$= 64 ({}^{n+1}C_0 8^{n-1} + \dots) + 8n + 8 - 8n - 9 \text{ have remainder } = 0$$

71. 2^{60} when divided by 7, leaves the remainder

- (1) 1 (2) 6 (3) 5 (4) 2

Sol. Answer (1)

$$\begin{aligned}(2^3)^{20} &= (7+1)^{20} = {}^{20}C_0 7^{20} + {}^{20}C_1 7^{19} + \dots + {}^{20}C_{19} 7 + {}^{20}C_{20} 7^0 \\ &= 7({}^{20}C_0 7^{19} + \dots + {}^{20}C_{19}) + 1\end{aligned}$$

Remainder = 1

72. The digit at unit's place in the number $17^{1995} + 11^{1995} - 7^{1995}$, is

- (1) 0 (2) 1 (3) 2 (4) 3

Sol. Answer (2)

$$(17)^{1995} + (11)^{1995} - 7^{1995}$$

Last digit of $(17)^{1995} = 7$

Last digit of $(11)^{1995} = 1$

Last digit of $(7)^{1995} = 7$

Last digit = 1

73. The coefficient of x^n in the polynomial

$(x + {}^nC_0)(x + 3 \cdot {}^nC_1)(x + 5 \cdot {}^nC_2)(x + 7 \cdot {}^nC_3) \dots (x + (2n+1) \cdot {}^nC_n)$ is

- (1) $n \cdot 2^{n-1}$ (2) $n \cdot 2^n$ (3) $n \cdot 2^{n+1}$ (4) $(n+1) 2^n$

Sol. Answer (4)

$$\begin{aligned}&(x + {}^nC_0)(x + 3 \cdot {}^nC_1)(x + 5 \cdot {}^nC_2) \dots (x + (2n+1) \cdot {}^nC_n) \\ &= x^{n+1} + ({}^nC_0 + 3 \cdot {}^nC_1 + 5 \cdot {}^nC_2 + \dots + (2n+1) \cdot {}^nC_n)x^n + \dots\end{aligned}$$

Coefficient of $x^n = {}^nC_0 + 3 \cdot {}^nC_1 + 5 \cdot {}^nC_2 + \dots + (2n+1) \cdot {}^nC_n$

$$= {}^nC_0 + (1+2) \cdot {}^nC_1 + (1+4) \cdot {}^nC_2 + \dots + (1+2n) \cdot {}^nC_n$$

$$= ({}^nC_0 + {}^nC_1 + \dots + {}^nC_n) + 2(1 \cdot {}^nC_1 + 2 \cdot {}^nC_2 + \dots + n \cdot {}^nC_n)$$

$$= 2^n + 2n \cdot 2^{n-1}$$

$$= 2^n + n \cdot 2^n$$

$$= (n+1) \cdot 2^n$$

74. Let $R = (5\sqrt{5} + 11)^{2n+1}$ and $f = R - [R]$ where $[\cdot]$ denotes the greatest integer function. The value of $R.f$ is

(1) 4^{2n+1}

(2) 4^{2n}

(3) 4^{2n-1}

(4) 4^{-2n}

Sol. Answer (1)

$$R = (5\sqrt{5} + 11)^{2n+1} = I + f$$

Where I is the integral part of R and f is the fractional part of R .

Hence $0 \leq f < 1$... (i)

Let us consider $f' = (5\sqrt{5} - 11)^{2n+1}$

Here also $0 < f' < 1$... (ii)

$$I + f = (5\sqrt{5} + 11)^{2n+1}$$

$$= {}^{2n+1}C_0(5)^{2n+1} + {}^{2n+1}C_1(5)^{2n}(11) + \dots + {}^{2n+1}C_{2n+1}(11)^{2n+1} \dots \text{(iii)}$$

$$f' = (5 - 11)^{2n+1}$$

$$= {}^{2n+1}C_0(5)^{2n+1} - {}^{2n+1}C_1(5)^{2n}(11) + \dots - (-1)^{2n+1} {}^{2n+1}C_{2n+1}(11)^{2n+1} \dots \text{(iv)}$$

Adding (iii) and (iv)

$$I + f - f' = 2[{}^{2n+1}C_1(5\sqrt{5})^{2n}(11) + {}^{2n+1}C_3(5\sqrt{5})^{2n-2}(11)^3 + \dots]$$

$$I + f - f' = \text{Even integer}$$

$$\Rightarrow I + f - f' = \text{Integer}$$

But I is an integer, hence $f - f'$ is an integer.

Also we know

$$0 \leq f < 1$$

$$0 < f' < 1$$

$$\Rightarrow -1 < -f' < 0$$

$$\text{Adding (v) and (vi)}$$

$$-1 < f - f' < 1$$

But $f - f'$ is an integer hence $f - f' = 0 \Rightarrow f = f'$

$$\Rightarrow Rf = Rf' = (5 + 11)^{2n+1} (5 - 11)^{2n+1} = (125 - 121)^{2n+1} = 4^{2n+1}$$

75. Consider the following statements

S_1 : The total number of terms in $(x^2 + 2x + 4)^{10}$ is 21

S_2 : The coefficient of x^{10} in $\left(x^2 + \frac{1}{x}\right)^{20}$ is ${}^{20}C_{10}$.

S_3 : The middle term in the expansion of $(1 + x)^{12}$ is ${}^{12}C_6 x^6$

S_4 : If the coefficients of fifth and ninth term in the expansion of $(1 + x)^n$ are same, then $n = 12$

Now identify the correct combination of true statements.

(1) S_1, S_2, S_3, S_4

(2) S_1, S_2 only

(3) S_2, S_3 only

(4) S_1, S_4 only

Sol. Answer (1)

$$S_1: (x^2 + 2x + 1 + 3)^{10} = [(x + 1)^2 + 3]^{10}$$

$$= {}^{10}C_0(x+1)^{20} + {}^{10}C_1(x+1)^{18} \cdot 3 + {}^{10}C_2(x+1)^{16} \cdot 3^2 + \dots + {}^{10}C_{10}3^{10}$$

Total number of different terms = 21

S_2 : Let $(r+1)^{\text{th}}$ term has x^{10}

$$\text{Then } T_{r+1} = {}^{20}C_r (x^2)^{20-r} \left(\frac{1}{x}\right)^r$$

$$= {}^{20}C_r x^{40-2r} \times \frac{1}{x^r}$$

$$x^{40-3r} = x^{10} \Rightarrow 40 - 3r = 10 \Rightarrow 30 = 3r \Rightarrow r = 10$$

Therefore coefficient of x^{10} is ${}^{20}C_{10}$.

S_3 : Total number of terms in the expansion of $(1+x)^{12}$ is 13

Therefore middle term is $\left(\frac{n}{2} + 1\right)^{\text{th}} = (6+1)^{\text{th}}$ term

$$T_{6+1} = {}^{12}C_6 x^6$$

$$S_4: T_{4+1} = T_{8+1}$$

$$\Rightarrow {}^nC_4 = {}^nC_8$$

$$\Rightarrow {}^nC_{n-4} = {}^nC_8$$

$$\Rightarrow n - 4 = 8$$

$$\Rightarrow n = 12$$

76. If $(1-x^3)^n = \sum_{r=0}^n a_r x^r (1-x)^{3n-2r}$, then the value of a_r where $n \in N$ is

(1) ${}^nC_r \cdot 3^r$

(2) ${}^nC_{3r}$

(3) ${}^nC_{r-1} 2^{r-1}$

(4) ${}^nC_r 2^r$

Sol. Answer (1)

$$\left\{ (1-x)(1+x+x^2) \right\}^n = \sum_{r=0}^n a_r \cdot x^r \cdot (1-x)^{3n-2r}$$

$$\Rightarrow (1+x+x^2)^n = \sum_{r=0}^n a_r \cdot x^r (1+x^2-2x)^{n-r}$$

$$\text{So, } \frac{a_r}{3^r} = {}^nC_r$$

$$\Rightarrow \boxed{a_r = {}^nC_r \cdot 3^r}$$

77. Let $(1+x^2)^2(1+x)^n = \sum_{k=0}^{n+4} a_k \cdot x^k$. If a_1, a_2 and a_3 are in A.P., then the value of n is

(1) 2, 3, 4

(2) 5, 6, 7

(3) 8, 9, 10

(4) -1, 4, 6

Sol. Answer (1)

$$(1+x^2)^2(1+x)^n = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots a_{n+4}x^{n+4}$$

$$\Rightarrow (1+x^2+2x^2)({}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \dots {}^nC_nx^n) = a_0 + a_1x + a_2x^2 + \dots \text{comparing the coefficients we find that}$$

$$a_1 = {}^nC_1 = n \quad \dots(i)$$

$$a_2 = {}^nC_2 + 2 \cdot {}^nC_0 = \frac{n(n-1)}{2} + 2$$

$$a_3 = {}^nC_3 + 2 \cdot {}^nC_1 = \frac{n(n-1)(n-2)}{6} + 2n$$

$$\text{But } 2a_2 = a_1 + a_3$$

$$\Rightarrow n(n-1)+4 = n + \frac{n(n-1)(n-2)}{6} + 2n$$

$$6n(n-1) + 24 = 6n + n(n-1)(n-2) + 12n$$

$$6n^2 - 6n + 24 = 18n + n(n^2 - 3n + 2)$$

$$n^3 - 3n^2 + 2n + 24n - 6n^2 - 24 = 0$$

$$n^3 - 9n^2 + 26n - 24 = 0$$

$$\Rightarrow n^2(n-2) - 7n(n-2) + 12(n-2) = 0$$

$$(n-2)(n^2 - 7n + 12) = 0$$

$$(n-2)(n-3)(n-4) = 0$$

$$\Rightarrow n = 2, 3, 4$$

78. The coefficient of x^{1007} in the expansion of

$$(1+x)^{2006} + x(1+x)^{2005} + x^2(1+x)^{2004} + x^3(1+x)^{2003} + \dots + x^{2006}$$

$$(1) {}^{2006}C_{1007}$$

$$(2) {}^{2006}C_{1006}$$

$$(3) {}^{2007}C_{1006}$$

$$(4) {}^{2007}C_{1007}$$

Sol. Answer (4)

$$\text{Let } S = (1+x)^{2006} + x(1+x)^{2005} + x^2(1+x)^{2004} + \dots + (1+x)^0 x^{2006}, \text{ it is a G.P.}$$

$$= \frac{(1+x)^{2006} \left[1 - \left(\frac{x}{1+x} \right)^{2007} \right]}{\left(1 - \frac{x}{1+x} \right)} = (1+x)^{2007} - x^{2007}$$

$$\text{Coefficient of } x^{1007} \text{ in } S = {}^{2007}C_{1007}$$

$$79. \sum_{r=0}^n (-1)^r \cdot {}^nC_r \left[\frac{1}{2^r} + \frac{3^r}{2^{2r}} + \frac{7^r}{2^{3r}} + \frac{15^r}{2^{4r}} + \dots m \text{ terms} \right] =$$

$$(1) \frac{2^{mn} - 1}{2^{mn}(2^n - 1)}$$

$$(2) \frac{2^{mn} - 1}{2^n - 1}$$

$$(3) \frac{2^{mn} + 1}{2^n + 1}$$

$$(4) \frac{2^{mn} + 1}{2^n - 1}$$

Sol. Answer (1)

$$\begin{aligned} & \sum_{r=0}^n (-1)^r \cdot {}^nC_r \left[\frac{1}{2^r} + \frac{3^r}{2^{2r}} + \frac{7^r}{2^{3r}} + \dots \text{upto } m \text{ terms} \right] \\ &= \sum_{r=0}^n (-1)^r \cdot {}^nC_r \left[\left(\frac{1}{2}\right)^r + \left(\frac{3}{4}\right)^r + \left(\frac{7}{8}\right)^r + \dots \text{upto } m \text{ terms} \right] \\ &= \sum_{r=0}^n (-1)^r \cdot {}^nC_r \left(\frac{1}{2}\right)^r + \sum_{r=0}^n (-1)^r \cdot {}^nC_r \left(\frac{3}{4}\right)^r + \sum_{r=0}^n {}^nC_r (-1)^r \left(\frac{7}{8}\right)^r + \dots \text{upto } m \text{ terms} \\ &= \left(1 - \frac{1}{2}\right)^n + \left(1 - \frac{3}{4}\right)^n + \left(1 - \frac{7}{8}\right)^n + \dots \text{upto } m \text{ terms} \\ &= \left(\frac{1}{2}\right)^n + \left(\frac{1}{4}\right)^n + \left(\frac{1}{8}\right)^n + \dots \text{upto } m \text{ terms} \\ &= \frac{\left(\frac{1}{2}\right)^n \left[\left(\frac{1}{2}\right)^{nm} - 1 \right]}{\left(\frac{1}{2}\right)^n - 1} = \frac{2^{mn} - 1}{2^{mn}(2^n - 1)} \end{aligned}$$

80. In the expansion of $(x + a)^n$, the sum of even terms is E , the sum of odd terms is O , then $O^2 - E^2$ is equal to

$$(1) (x^2 + a^2)^n$$

$$(2) (x^2 - a^2)^n$$

$$(3) (x^2 - a^2)^{2n}$$

$$(4) (x^2 + a^2)^{2n}$$

Sol. Answer (2)

$$\begin{aligned} (x + a)^n &= {}^nC_0 x^n a^0 + {}^nC_1 x^{n-1} a + {}^nC_2 x^{n-2} a^2 + \dots + {}^nC_n a^n \\ &= T_1 + T_2 + T_3 + \dots + T_{n+1} \\ &= (T_1 + T_3 + T_5 + \dots) + (T_2 + T_4 + T_6 + \dots) \end{aligned}$$

$$(x + a)^n = O + E \quad \dots (i)$$

$$\text{Similarly } (x - a)^n = O - E \quad \dots (ii)$$

Multiplying (i) and (ii)

$$(x^2 - a^2)^n = O^2 - E^2$$

81. For $4 \leq r \leq n$,

$$\binom{n}{r} + 4\binom{n}{r+1} + 6\binom{n}{r+2} + 4\binom{n}{r+3} + \binom{n}{r+4} \text{ equals}$$

(1) $\binom{n+4}{r+4}$

(2) $\binom{n+4}{r}$

(3) $\binom{n+3}{r-1}$

(4) $\binom{n+4}{r+3}$

Sol. Answer (1)

$$\begin{aligned} & \binom{n}{r} + 4\binom{n}{r+1} + 6\binom{n}{r+2} + 4\binom{n}{r+3} + \binom{n}{r+4} \\ &= {}^nC_r + 4 \cdot {}^nC_{r+1} + 6 \cdot {}^nC_{r+2} + 4 \cdot {}^nC_{r+3} + {}^nC_{r+4} \\ &= ({}^nC_r + {}^nC_{r+1}) + 3({}^nC_{r+1} + {}^nC_{r+2}) + 3({}^nC_{r+2} + {}^nC_{r+3}) + ({}^nC_{r+3} + {}^nC_{r+4}) \\ &= ({}^{n+1}C_{r+1}) + 3({}^{n+1}C_{r+2}) + 3({}^{n+1}C_{r+3}) + ({}^{n+1}C_{r+4}) \\ &= ({}^{n+1}C_{r+1} + {}^{n+1}C_{r+2}) + 2({}^{n+1}C_{r+2} + {}^{n+1}C_{r+3}) + ({}^{n+1}C_{r+3} + {}^{n+1}C_{r+4}) \\ &= {}^{n+2}C_{r+2} + 2({}^{n+2}C_{r+3}) + {}^{n+2}C_{r+4} \\ &= ({}^{n+2}C_{r+2} + {}^{n+2}C_{r+3}) + ({}^{n+2}C_{r+3} + {}^{n+2}C_{r+4}) \\ &= {}^{n+3}C_{r+3} + {}^{n+3}C_{r+4} \\ &= {}^{n+4}C_{r+4} = \binom{n+4}{r+4} \end{aligned}$$

82. If $\sum_{k=0}^n (k^2 + k + 1)k! = (2007) \cdot 2007!$, then value of n is

(1) 2007

(2) 2006

(3) 2008

(4) 2005

Sol. Answer (2)

$$\sum_{k=0}^n (k^2 + k + 1)k! \equiv 2007 \cdot 2007!$$

$$\text{But } \sum_{k=0}^n (k^2 + k + 1)k! = \sum_{k=0}^n (k^2 + k + 1)k!$$

$$= \sum_{k=0}^n (k^2 \cdot k! + (k+1)!) = \sum_{k=0}^n ((k^2 - 1 + 1)k! + (k+1)!)$$

$$= \sum_{k=0}^n [(k-1)(k+1) + 1]k! + (k+1)! = \sum_{k=0}^n [(k-1)(k+1)! + k! + (k+1)!]$$

$$\begin{aligned}
&= \sum_{k=0}^n (k+1)! k + k! = \sum_{k=0}^n [(k+1)! (k+1-1) + k!] \\
&= \sum_{k=0}^n [(k+1)! (k+1) + [k! - (k+1)!]] = \sum_{k=0}^n [(k+2)! - (k+1)!] + [k! - (k+1)!] \\
&= \sum_{k=0}^n [(k+2)! - (k+1)!] - [(k+1)! - k!] \\
&= [(2! - 1!) - (1! - 0!)] + [(3! - 2!) - (2! - 1!)] + \dots + [((n+2)! - (n+1)!) - ((n+1)! - n!)] \\
&= [(2! - 1!) + (3! - 2!) + \dots + ((n+2)! - (n+1)!)] - [(1! - 0!) + (2! - 1!) + (3! - 2!) + \dots + ((n+1)! - n!)] \\
&= [(n+2)! - 1!] - [(n+1)! - 0!] \\
&= (n+2)! - (n+1)! = (n+1)! (n+1) = (2007)! (2007!)
\end{aligned}$$

Comparing $n = 2006$

83. There are two bags each of which contains n balls. A man has to select an equal number of balls from both the bags. Find the number of ways in which man can choose at least one ball from each bag.

(1) ${}^{2n}C_n$ (2) ${}^{2n}C_n - 1$ (3) $n!$ (4) ${}^{2n}C_n - 2$

Sol. Answer (2)

Required number of ways

$$\begin{aligned}
&= {}^nC_1 \cdot {}^nC_1 + {}^nC_2 \cdot {}^nC_2 + \dots + {}^nC_n \cdot {}^nC_n - {}^{2n}C_n - 1 \\
&= 0
\end{aligned}$$

84. The value of $C_0 \cdot {}^{2n}C_n - C_1 \cdot {}^{2n-2}C_n + C_2 \cdot {}^{2n-4}C_n - \dots$ is

(1) 2^n (2) n^2 (3) 0 (4) n

Sol. Answer (1)

$$\begin{aligned}
&{}^nC_0 \cdot {}^{2n}C_n - {}^nC_1 \cdot {}^{2n-2}C_n + {}^nC_2 \cdot {}^{2n-4}C_n \\
&((1+x)^2 - 1)^n = {}^nC_0 (1+x)^{2n} - {}^nC_1 (1+x)^{2(n-1)} \cdot x + {}^nC_2 (1+x)^{2(n-2)} \cdot x^2 \\
&(x^2 + 2x)^n = {}^nC_0 (1+x)^{2n} - {}^nC_1 (1+x)^{2(n-1)} x + \dots
\end{aligned}$$

Equating both side coefficient of x^n

$$2^n = {}^nC_0 \cdot {}^{2n}C_n - {}^nC_1 \cdot {}^{2n-2}C_n + \dots$$

85. The value of $(C_0 + C_1)(C_1 + C_2) \dots (C_{n-1} + C_n)$ is

(1) $\frac{(n+1)^n}{n!} \cdot C_1 C_2 \dots C_n$ (2) $\frac{(n-1)^n}{n!} \cdot C_1 C_2 \dots C_n$

(3) $\frac{(n)^n}{(n+1)!} \cdot C_1 C_2 \dots C_n$ (4) $\frac{(n)^n}{n!} \cdot C_1 C_2 \dots C_n$

Sol. Answer (1)

$$\begin{aligned}
 & (C_0 + C_1)(C_1 + C_2) \dots (C_{n-1} + C_n) \\
 & {}^{n+1}C_1 \cdot {}^{n+1}C_2 \dots {}^{n+1}C_n \\
 & = \frac{1 \cdot {}^{n+1}C_1 \cdot 2 \cdot {}^{n+1}C_2 \cdot 3 \cdot {}^{n+1}C_3 \dots n \cdot {}^{n+1}C_n}{1 \cdot 2 \cdot 3 \dots n} \\
 & = \frac{(n+1)^n \cdot {}^nC_0 \cdot {}^nC_1 \dots {}^nC_n}{n!} \\
 & = \frac{(n+1)^n}{n!} \cdot C_1 \cdot C_2 \cdot C_3 \dots C_n
 \end{aligned}$$

86. For $r = 0, 1, \dots, 10$, let A_r , B_r and C_r denote, respectively, the coefficient of x^r in the expansions of $(1+x)^{10}$, $(1+x)^{20}$ and $(1+x)^{30}$. Then $\sum_{r=1}^{10} A_r (B_{10} B_r - C_{10} A_r)$ is equal to **[IIT-JEE 2010]**

- (1) $B_{10} - C_{10}$ (2) $A_{10} (B_{10}^2 - C_{10} A_{10})$ (3) 0 (4) $C_{10} - B_{10}$

Sol. Answer (4)

$$\begin{aligned}
 A_r &= \text{Coefficient of } x^r \text{ in } (1+x)^{10} = {}^{10}C_r \\
 B_r &= \text{Coefficient of } x^r \text{ in } (1+x)^{20} = {}^{20}C_r \\
 C_r &= \text{Coefficient of } x^r \text{ in } (1+x)^{30} = {}^{30}C_r \\
 \sum_{r=1}^{10} A_r (B_{10} B_r - C_{10} A_r) &= \sum_{r=1}^{10} A_r B_{10} B_r - \sum_{r=1}^{10} A_r C_{10} A_r \\
 &= \sum_{r=1}^{10} {}^{10}C_r {}^{20}C_{10} {}^{20}C_r - \sum_{r=1}^{10} {}^{10}C_r {}^{30}C_{10} {}^{10}C_r \\
 &= \sum_{r=1}^{10} {}^{10}C_{10-r} {}^{20}C_{10} {}^{20}C_r - \sum_{r=1}^{10} {}^{10}C_{10-r} {}^{30}C_{10} {}^{10}C_r \\
 &= {}^{20}C_{10} \sum_{r=1}^{10} {}^{10}C_{10-r} {}^{20}C_r - {}^{30}C_{10} \sum_{r=1}^{10} {}^{10}C_{10-r} {}^{10}C_r \\
 &= {}^{20}C_{10} ({}^{30}C_{10} - 1) - {}^{30}C_{10} ({}^{20}C_{10} - 1) \\
 &= {}^{30}C_{10} - {}^{20}C_{10} = C_{10} - B_{10}
 \end{aligned}$$

87. Coefficient of x^{11} in the expansion of $(1 + x^2)^4 (1 + x^3)^7 (1 + x^4)^{12}$ is

[JEE(Advanced)-2014]

(1) 1051

(2) 1106

(3) 1113

(4) 1120

Sol. Answer (3)

Power of

Coefficient of x^{11} x^2 x^3 x^4

$${}^4C_0 \times {}^7C_1 \times {}^{12}C_2$$

0 1 2

$${}^4C_2 \times {}^7C_1 \times {}^{12}C_1$$

2 1 1

$${}^4C_4 \times {}^7C_1 \times {}^{12}C_0$$

4 1 0

$${}^4C_1 \times {}^7C_3 \times {}^{12}C_0$$

1 3 0

1113

SECTION - B

Objective Type Questions (More than one options are correct)

1. For a positive integer n , if the expansion of $\left(\frac{5}{x^2} + x^4\right)^n$ has a term independent of x , then n can be

(1) 18

(2) 21

(3) 27

(4) 99

Sol. Answer (1, 2, 3, 4)

$$T_{r+1} = {}^nC_r \left(\frac{5}{x^2}\right)^{n-r} (x^4)^r = {}^nC_r 5^{n-r} x^{-2n+2r} x^{4r} = {}^nC_r 5^{n-r} x^{-2n+6r}$$

For term independent of 'x'

$$-2n + 6r = 0$$

$$\Rightarrow n = 3r = 0, 3, 6, 9, \dots$$

2. The positive value of a so that the coefficients of x^5 is equal to that of x^{15} in the expansion of $\left(x^2 + \frac{a}{x^3}\right)^{10}$ is(1) $\frac{1}{2\sqrt{3}}$ (2) $\frac{1}{\sqrt{3}}$ (3) $\frac{\sqrt{3}}{6}$ (4) $\frac{1}{3}$

Sol. Answer (1, 3)

$$\left(x^2 + \frac{a}{x^3}\right)^{10}$$

$$T_{r+1} = {}^{10}C_r (x^2)^{10-r} \left(\frac{a}{x^3}\right)^r$$

$$= {}^{10}C_r a^r x^{20-2r} x^{-3r}$$

$$= {}^{10}C_r a^r x^{20-5r}$$

$$\text{If } 20 - 5r = 5 \Rightarrow r = 3$$

$$\Rightarrow \text{Coefficient of } x^5 = {}^{10}C_3 a^3$$

$$\text{If } 20 - 5r = 15 \Rightarrow r = 1$$

$$\Rightarrow \text{Coefficient of } x^{15} = {}^{10}C_1 a^1 = {}^{10}C_1 a$$

$$\text{Given that } {}^{10}C_3 a^3 = {}^{10}C_1 a^1$$

$$\Rightarrow \frac{10 \times 9 \times 8}{3 \times 2} a^3 = 10 \times a$$

$$\Rightarrow a^2 = \frac{1}{12}$$

$$\Rightarrow a = \frac{1}{2\sqrt{3}} = \frac{\sqrt{3}}{6}$$

$$\text{Positive value of } a = \frac{1}{2\sqrt{3}} \text{ or } \frac{\sqrt{3}}{6}$$

3. The sum of the coefficients of all the even powers of x in the expansion of $(2x^2 - 3x + 1)^{11}$ is

(1) 3.6^{10}

(2) 6^{11}

(3) $2^{10}.3^{11}$

(4) $2^{11}.3^{10}$

Sol. Answer (1, 3)

$$(2x^2 - 3x + 1)^{11} = A_0 + A_1x + A_2x^2 + \dots + A_{22}x^{22}$$

Putting $x = 1$, we get

$$0 = A_0 + A_1 + A_2 + \dots + A_{22} \quad \dots (i)$$

Putting $x = -1$, we get

$$6^{11} = A_0 - A_1 + A_2 - A_3 + \dots + A_{22} \quad \dots (ii)$$

Adding equation (i) and (ii)

$$6^{11} = 2[A_0 + A_2 + A_4 + \dots]$$

Sum of coefficient of even powers of x

$$= A_0 + A_2 + A_4 + \dots$$

$$= \frac{6^{11}}{2} = \frac{6.6^{10}}{2}$$

$$= 3.6^{10} = 3.2^{10}.3^{10} = 2^{10}.3^{11}$$

4. The term independent of x in the expansion of $\left(x - \frac{1}{x}\right)^4 \left(x + \frac{1}{x}\right)^3$ is

(1) -3

(2) 0

(3) nP_k , where $k > n$

(4) nC_r , where $r > n$

Sol. Answer (2, 3, 4)

$$\left(x - \frac{1}{x}\right)^4 \left(x + \frac{1}{x}\right)^3$$

$$= \left(x - \frac{1}{x}\right) \left(x^2 - \frac{1}{x^2}\right)^3$$

$$= \left(x - \frac{1}{x}\right) \left(x^6 + \frac{1}{x^6} - 3\left(x^2 - \frac{1}{x^2}\right)\right)$$

$$= \left(x - \frac{1}{x}\right) \left(x^6 + \frac{1}{x^6} - 3x^2 + \frac{3}{x^2}\right)$$

Clearly no term is independent of x

Hence coefficient = 0

Also coefficient = nC_k if $k > n$

$$= {}^nP_k \text{ if } k > n$$

Hence, options, 2, 3, 4 are correct.

5. If the second, third and fourth terms in the expansion of $(x + y)^n$ be 135, 30 and $10/3$ respectively, then

$$(1) \ x = 3$$

$$(2) \ y = \frac{1}{3}$$

$$(3) \ n = 5$$

$$(4) \ n = 7$$

Sol. Answer (1, 2, 3)

$$T_2 = {}^nC_1 x^{n-1} y = 135 \quad \dots(i)$$

$$T_3 = {}^nC_2 x^{n-2} y^2 = 30 \quad \dots(ii)$$

$$T_4 = {}^nC_3 x^{n-3} y^3 = \frac{10}{3} \quad \dots(iii)$$

If (ii) is divided by (i)

$$\text{Then, we get } \left(\frac{n-1}{2}\right) \left(\frac{y}{x}\right) = \frac{30}{135} \quad \dots(iv)$$

By dividing equation (iii) by (ii)

$$\left(\frac{n-2}{3}\right) \left(\frac{y}{x}\right) = \frac{10}{3 \times 30} = \frac{1}{9} \quad \dots(v)$$

$$\frac{\left(\frac{n-1}{2}\right)}{\left(\frac{n-2}{3}\right)} = \frac{\left(\frac{30}{135}\right)}{\left(\frac{1}{9}\right)}$$

by (iv) and (v)

$$\frac{3(n-1)}{2(n-2)} = \frac{9 \times 30}{135}$$

$$\frac{3(n-1)}{2(n-2)} = \frac{30}{15} = 2$$

$$3n - 3 = 4n - 8$$

$$n = 5$$

$$\text{By (iv)} \left(\frac{5-1}{2}\right)\left(\frac{y}{x}\right) = \frac{30}{135} = \frac{2}{9}$$

$$2\left(\frac{y}{x}\right) = \frac{2}{9}$$

$$\frac{y}{x} = \frac{1}{9} \text{ (iv)} \Rightarrow y = \frac{x}{9}$$

$$\text{By eqn. (i)} {}^5C_1 x^4 \left(\frac{x}{9}\right) = 135$$

$$\Rightarrow x^5 = 3^5 \Rightarrow x = 3 \text{ and } y = \frac{1}{3}$$

6. $\sum_{r=0}^6 (-1)^r \binom{16}{r}$ is divisible by

(1) 5

(2) 7

(3) 11

(4) 13

Sol. Answer (1, 2, 3, 4)

$$\sum_{r=0}^{16} (-1)^r {}^{16}C_r = {}^{16}C_0 - {}^{16}C_1 + {}^{16}C_2 - {}^{16}C_3 + \dots - {}^{16}C_{16} = 0$$

Hence, the number is divisible by all 5, 7, 11, 13

Hence, all options are correct.

7. If $(1 + 2x + 3x^2)^{10} = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots + a_{20}x^{20}$, then

(1) $a_1 = 20$

(2) $a_2 = 210$

(3) $a_4 = 8085$

(4) $a_{20} = 2^2 \cdot 3^7 \cdot 7$

Sol. Answer (1, 2, 3)

$$(1 + 2x + 3x^2)^{10} = 1 + {}^{10}C_1(2x + 3x^2) + {}^{10}C_2(2x + 3x^2)^2 + {}^{10}C_3(2x + 3x^2)^3 + {}^{10}C_4(2x + 3x^2)^4 + \dots$$

$$= 1 + x(2 \cdot {}^{10}C_1) + x^2(3 \cdot {}^{10}C_1 + 4 \cdot {}^{10}C_2) + x^3(12 \cdot {}^{10}C_2 + 8) + x^4(9 \cdot {}^{10}C_2 + 3 \cdot 2 \cdot 3 \cdot 2 \cdot {}^{10}C_3 + {}^{10}C_4 \cdot 2^4) + n^4(\dots) + \dots$$

Comparing the coefficients

$$a_0 = 1, a_1 = 2 \times 10 = 20$$

$$a_2 = 30 + \frac{4 \times 10 \times 9}{2} = 30 + 180 = 210$$

$$a_4 = 9 \cdot {}^{10}C_2 + 36 \cdot {}^{10}C_3 + {}^{10}C_4 \cdot 16$$

$$= 9 \cdot \frac{10 \times 9}{2} + 36 \times \frac{10 \times 9 \times 8}{3 \times 2} + \frac{10 \times 9 \times 8 \times 7}{4 \times 3 \times 2 \times 1} \cdot 16$$

$$= 8085$$

8. The maximum value of nC_r is obtained when r is equal to

- (1) $\frac{n}{3}$ (2) $\frac{n}{4}$ (3) $\frac{n-1}{2}$ or $\frac{n+1}{2}$ (4) $\frac{n}{2}$

Sol. Answer (3, 4)

$${}^nC_r \text{ is maximum if } r = \begin{cases} \frac{n}{2} & ; \text{ if } n \text{ is even} \\ \frac{n-1}{2} \text{ or } \frac{n+1}{2} & ; \text{ if } n \text{ is odd} \end{cases}$$

Hence options (3), (4) are correct.

9. Given that the 4th term in the expansion of $\left(2 + \frac{3x}{8}\right)^{10}$ has the maximum numerical value, then x lies in the interval

- (1) $\left(2, \frac{64}{21}\right)$ (2) $\left(-\frac{60}{23}, -2\right)$ (3) $\left(-\frac{64}{21}, -2\right)$ (4) $\left(2, -\frac{60}{23}\right)$

Sol. Answer (1, 3)

If 4th term has maximum numerical value then

$$|T_5| < |T_4| > |T_3|$$

(i) Using $|T_4| > |T_3|$

$$\left|\frac{T_4}{T_3}\right| > 1$$

$$\Rightarrow \left|\frac{{}^{10}C_3(2)^7\left(\frac{3x}{8}\right)^3}{{}^{10}C_2(2)^8\left(\frac{3x}{8}\right)^2}\right| > 1 \Rightarrow \left(\frac{10-3+1}{3}\right)\left|\frac{3x}{8} \times \frac{1}{2}\right| > 1$$

$$\Rightarrow \left|\frac{8}{3}\right|\left|\frac{3}{8} \cdot \frac{x}{2}\right| > 1$$

$$\Rightarrow |x| > 2 \quad \Rightarrow x \in (-\infty, -2) \cup (2, \infty) \quad \dots (i)$$

(ii) Using $|T_4| > |T_5|$

$$|T_5| < |T_4|$$

$$\left|\frac{T_5}{T_4}\right| < 1$$

$$\Rightarrow \left|\left(\frac{10-4+1}{4}\right)\frac{\left(\frac{3x}{8}\right)}{2}\right| < 1 \Rightarrow \left|\frac{7}{4} \cdot \frac{3x}{8} \times \frac{1}{2}\right| < 1$$

$$\Rightarrow |x| < \frac{64}{21}$$

$$\Rightarrow -\frac{64}{21} < x < \frac{64}{21} \quad \dots (ii)$$

By equation (i) and (ii)

$$x \in \left(2, \frac{64}{21}\right) \text{ or } \left(-\frac{64}{21}, -2\right)$$

10. The value of the expression $C_0^2 - C_1^2 + C_2^2 - \dots + (-1)^n C_n^2$ is

(1) 0 if n is odd

(2) $(-1)^n$ if n is odd

(3) $(-1)^{n/2} {}^n C_{n/2}$ if n is even

(4) $(-1)^{n-1} {}^n C_{n-1}$ if n is even

Sol. Answer (1, 3)

$$(x+1)^n = C_0 x^n + C_1 x^{n-1} + C_2 x^{n-2} + \dots + C_{n-1} x + C_n \quad \dots (i)$$

$$(1-x)^n = C_0 - C_1 x + C_2 x^2 - C_3 x^3 + \dots + (-1)^n C_n x^n \quad \dots (ii)$$

Multiply in equation (i) and (ii)

$$(1-x^2)^n = (C_0^2 - C_1^2 + C_2^2 - C_3^2 + \dots + (-1)^n C_n^2) x^n + \dots$$

Hence $C_0^2 - C_1^2 + C_2^2 - C_3^2 + \dots + (-1)^n C_n^2 = \text{Coefficient of } x^n \text{ in } (1-x^2)^n$

$$= \begin{cases} 0 & ; \text{if } n \text{ is odd} \\ (-1)^{\frac{n}{2}} {}^n C_{\frac{n}{2}} & ; \text{if } n \text{ is even} \end{cases}$$

11. The number $101^{100} - 1$ is divisible by

(1) 10^2

(2) 10^3

(3) 10^4

(4) 10^5

Sol. Answer (1, 2, 3)

$$\left((101)^{100} - 1\right) = (1+100)^{100} - 1$$

$$= \left(1 + {}^{100}C_1 100 + {}^{100}C_2 (100)^2 + \dots + {}^{100}C_{100} (100)^{100}\right) - 1$$

$$= 10^4 (1 + {}^{10}C_4 + \dots)$$

$$= 10^4 \times \text{integer}$$

Hence, $(101)^{100} - 1$ is divided by 10^4 and hence divided by 10^3 and 10^2 also.

12. If n is a positive integer and $(3\sqrt{3}+5)^{2n+1} = I + f$ where I is an integer and $0 < f < 1$, then

(1) I is an even integer

(2) $(I+f)f$ is divisible by 2^{2n+1}

(3) The integer just less than $(3\sqrt{3}+5)^{2n+1}$ is divisible by 3

(4) I is divisible by 10

Sol. Answer (1, 2, 4)

$$(3\sqrt{3} + 5)^{2n+1} = I + f \text{ where } I \text{ is an integer and } 0 < f < 1$$

$$\text{Let us take } f' = (3\sqrt{3} - 5)^{2n+1} \text{ where } 0 < f' < 1$$

Here for solutions we expand both $I + f$ and f'

$$\begin{aligned} (I + f) &= (3\sqrt{3} + 5)^{2n+1} \\ &= {}^{2n+1}C_0 (3\sqrt{3})^{2n+1} + {}^{2n+1}C_1 (3\sqrt{3})^{2n} (5) + \dots + {}^{2n+1}C_{2n+1} 5^{2n+1} \end{aligned} \quad \dots(i)$$

$$\begin{aligned} f' &= (3\sqrt{3} - 5)^{2n+1} \\ &= {}^{2n+1}C_0 (3\sqrt{3})^{2n+1} - {}^{2n+1}C_1 (3\sqrt{3})^{2n} (5) + \dots + {}^{2n+1}C_{2n+1} 5^{2n+1} \end{aligned} \quad \dots(ii)$$

By (i) and (ii)

$$I + f - f' = 2 \left[{}^{2n+1}C_1 (3\sqrt{3})^{2n} (5) + {}^{2n+1}C_3 (3\sqrt{3})^{2n-2} (5)^3 + \dots + 5^{2n+1} \right] \quad \dots (iii)$$

$$= 2 \times \text{an integer} = \text{even integer}$$

$$\Rightarrow I + f - f' = \text{an integer}$$

$$\Rightarrow f - f' \text{ is an integer}$$

$$\text{Again } 0 < f < 1$$

$$0 < f' < 1$$

$$\Rightarrow -1 < f' < 0$$

$$\text{Adding (iv) and (v)}$$

$$-1 < f - f' < 1, \text{ but } f - f' \text{ is an integer hence}$$

$$f - f' = 0 \quad \Rightarrow \quad f = f'$$

$$\text{As } f - f' = 0 \text{ and } I + f - f' = \text{even integer}$$

$$\Rightarrow I \text{ is an even integer}$$

$$\text{By (iii) } I = 2 \times 5 \times \text{Integer}$$

$$I = 10 \times \text{integer, hence } I \text{ is divisible by } 10$$

If an integer just less than $(3\sqrt{3} + 5)^{2n+1}$ is I which is an even integer hence it may or may not be divisible by 3.

$$\begin{aligned} (I + f)f &= (3\sqrt{3} + 5)^{2n+1} (3\sqrt{3} - 5)^{2n+1} \\ &= (27 - 25)^{2n+1} = 2^{2n+1} \end{aligned}$$

$$\text{Hence } (I + f)f \text{ is divisible by } 2^{2n+1}$$

13. If n is a positive integer and if $(1+x+x^2)^n = \sum_{r=0}^{2n} a_r x^r$, then

$$(1) a_r = a_{2n-r}, \text{ for } 0 \leq r \leq 2n$$

$$(2) a_0 + a_1 + \dots + a_{n-1} = \frac{1}{2}(3^n - a_n)$$

$$(3) a_0^2 - a_1^2 + a_2^2 - a_3^2 + \dots + a_{2n}^2 = a_n$$

$$(4) a_0 + a_2 + \dots + a_{2n} = \frac{1}{2}(3^n + 1)$$

Sol. Answer (1, 2, 3, 4)

We have

$$\sum_{r=0}^{2n} a_r \cdot x^r = (1+x+x^2)^n$$

Replacing x by $\frac{1}{x}$ on both sides,

$$\sum_{r=0}^{2n} a_r \cdot \left(\frac{1}{x}\right)^r = \left(1 + \frac{1}{x} + \frac{1}{x^2}\right)^n$$

$$= \left(\frac{x^2+x+1}{x^2}\right)^n$$

$$= (x^2+x+1)^n \cdot \frac{1}{x^{2n}}$$

$$\Rightarrow \sum_{r=0}^{2n} a_r \cdot x^{2n-r} = (x^2+x+1)^n$$

$$= \sum_{r=0}^{2n} a_r x^r$$

$$= \sum_{r=0}^{2n} a_{2n-r} x^{2n-r} \quad \dots(i)$$

Comparing both sides we get,

$$a_r = a_{2n-r} \text{ for } 0 \leq r \leq 2n$$

Again we have,

$$a_0 + a_1 x + a_2 x^2 + \dots + a_{2n} x^{2n} = (1+x+x^2)^n$$

Putting $x = 1$ both sides we get,

$$a_0 + a_1 + a_2 + \dots + a_{2n} = 3^n.$$

$$\text{But } a_{-r} = a_{2n-r}$$

$$2(a_0 + a_1 + a_2 + \dots + a_{n-1}) + a_n = 3^n$$

$$\Rightarrow a_0 + a_1 + a_2 + \dots + a_{n-1} = \frac{1}{2} [3^n - a_n]$$

Again,

$$(1 + x + x^2)^n = a_0 + a_1x + a_2x^2 + \dots + a_{2n}x^{2n} \quad \dots(ii)$$

Replacing x by $-\frac{1}{x}$, we get

$$\left(1 - \frac{1}{x} + \frac{1}{x^2}\right)^n = a_0 - \frac{a_1}{x} + \frac{a_2}{x^2} + \frac{a_3}{x^3} + \dots + \frac{a_{2n}}{x^{2n}} \quad \dots(iii)$$

Multiplying (ii) and (iii), comparing constant term we get,

$$\begin{aligned} a_0^2 - a_1^2 + a_2^2 - a_3^2 + \dots + a_{2n}^2 &= a_n \\ a_0 - a_1 + a_2 - a_3 + \dots + a_{2n} &= 1 \\ a_0 + 0a_1 + a_2 - a_3 + \dots + a_{2n} &= 3^n \\ \hline a_0 + a_2 + \dots + a_{2n} &= \frac{1}{2}(3^n + 1) \end{aligned}$$

Hence (1), (2), (3), (4) are correct.

14. Which of the following is/are correct?

$$(1) \quad (101^{50} - 99^{50}) > 100^{50}$$

$$(2) \quad (101)^{50} - 100^{50} > 99^{50}$$

$$(3) \quad (1000)^{1000} > (1001)^{999}$$

$$(4) \quad (1001)^{999} > (1000)^{1000}$$

Sol. Answer (1, 2, 3)

$$\begin{aligned} 101^{50} - 99^{50} &= (100 + 1)^{50} - (100 - 1)^{50} \\ &= [{}^{50}C_0 \cdot (100)^{50} \cdot 1^{50} + {}^{50}C_1 \cdot (100)^{49} \cdot 1^1 + {}^{50}C_2 \cdot (100)^{48} \cdot 1^2 + \dots] \\ &\quad - [{}^{50}C_0 \cdot (100)^{50} \cdot (-1)^0 - {}^{50}C_1 \cdot (100)^{49} \cdot 1^1 + {}^{50}C_2 \cdot (100)^{48} \cdot 1^2 + \dots] \\ &= (100)^{50} + 2[{}^{50}C_3(100)^{47} + {}^{50}C_5(100)^{45} + \dots] > (100)^{50} \\ \Rightarrow (101)^{50} - (100)^{50} &> 99^{50} \end{aligned}$$

$$\text{Also, } \left(\frac{1005}{1000}\right)^{999} = \left(1 + \frac{1}{1000}\right)^{999}$$

$$= 1 + {}^{999}C_1 \cdot \frac{1}{1000} + {}^{999}C_2 \cdot \left(\frac{1}{1000}\right)^2 + \dots < 1 + 1 + \dots + 1 = 1000$$

$$\therefore \left(\frac{1001}{1000}\right)^{999} < 1000$$

$$\Rightarrow (1001)^{999} < (1000)^{1000}$$

Hence option (1), (2), (3) are correct.

15. If C_r stands for ${}^nC_r = \frac{n!}{r! (n-r)!}$ and $\sum_{r=1}^n r.C_r^2 = \lambda$ for $n \geq 2$, then λ is divisible by

- (1) $3(n-1)$ (2) $n+1$ (3) $n(2n-1)$ (4) n^2+1

Sol. Answer (3)

$$(x+1)^n = {}^nC_0 x^n + {}^nC_1 x^{n-1} + \dots + {}^nC_n 1^n \quad \dots (i)$$

$$\text{or } (x+1)^n = C_0 x^n + C_1 x^{n-1} + \dots + C_n \quad \dots (ii)$$

where $C_r = {}^nC_r$

$$\text{Again } (1+x)^n = C_0 + C_1 x + C_2 x^2 + \dots + C_n x^n$$

Differentiating

$$n(1+x)^{n-1} = C_1 + 2C_2 x + 3C_3 x^2 + \dots + nC_n x^{n-1} \quad \dots (iii)$$

Multiplying (ii) and (iii), we get

$$n(1+x)^{2n-1} = (C_0 x^n + C_1 x^{n-1} + C_2 x^{n-2} + \dots + C_n)(C_1 + 2C_2 x + 3C_3 x^2 + \dots + nC_n x^{n-1})$$

Comparing the coefficient of x^{n-1}

$$n \cdot {}^{2n-1}C_{n-1} = C_1^2 + 2C_2^2 + 3C_3^2 + \dots + nC_n^2$$

$$\Rightarrow C_1^2 + 2C_2^2 + 3C_3^2 + \dots + nC_n^2 = n \cdot \frac{(2n-1)!}{(n-1)! n!}$$

which is clearly divisible by $n(2n-1)$

SECTION - C

Linked Comprehension Type Questions

Comprehension-I

If $S = 1! + 4! + 7! + 10! + \dots + 400!$, then

1. The ten's digit in the number S is divisible by

- (1) 4 (2) 6 (3) 5 (4) 7

Sol. Answer (2)

2. The last two digits in the number S is divisible by

- (1) 13 (2) 12 (3) 11 (4) 10

Sol. Answer (1)

3. The last two digits in $(1! + 4! + 7!)!$ is

- (1) 01 (2) 10 (3) 00 (4) None of these

Sol. Answer (3)

$$S = 1! + 4! + 7! + \dots 400!$$

$$S = 1! + 4! + 7! + 10! + 13! + \dots + 400!$$

As $10!$, $11!$, $12!$, all have '00' at the end

and $5!$, $6!$, $7!$, ... all have '0' at the unit place

Hence last two digit is decided by only the number $1! + 4! + 7!$

$$\text{and } 1! + 4! + 7! = 1 + 24 + 5040$$

$$= 5065$$

Hence last two digits in S is 65. Digit at ten's place in S is 6

65 is divisible by 13

Last two digits in $(1! + 4! + 7!)!$ or $(5065)!$ is 00.

Comprehension-II

If $C_0, C_1, C_2, C_3, \dots, C_n$ be binomial coefficients in the expansion of $(1+x)^n$, then

- The value of the expression $C_0 + 2C_1 + 3C_2 + \dots + (n+1)C_n$ is equal to
 (1) $2^{n-1}(n+1)$ (2) $2^{n-1}(n+2)$ (3) $2^n(n+2)$ (4) None of these
- The value of the expression $C_0 - 2C_1 + 3C_2 - \dots + (-1)^n(n+1)C_n$ is equal to
 (1) 0 (2) $2^n(n+3)$ (3) $2^{n-1}(n-2)$ (4) None of these
- The value of $C_0 + (C_0 + C_1) + (C_0 + C_1 + C_2) + \dots + (C_0 + C_1 + C_2 + \dots + C_{n-1})$ equal
 (1) $n2^n$ (2) $n2^{n-1}$ (3) $(n+2)2^n$ (4) None of these

Solution of Comprehension-II

$$(1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n$$

$$x(1+x)^n = C_0x + C_1x^2 + C_2x^3 + \dots + C_nx^{n+1}$$

Differentiating

$$x.n(1+x)^{n-1} + (1+x)^n = C_0 + 2C_1x + 3C_2x^2 + \dots + n.C_nx^{n-1} \quad \dots(i)$$

By putting $x = 1$, in (i) we get

$$C_0 + 2C_1 + 3C_2 + 4C_3 + \dots = 2^{n-1}(n+2)$$

Putting $x = -1$, in (i)

$$0 = C_0 - 2C_1 + 3C_2 + \dots (-1)^n(n+1)C_n$$

- Answer (2)

$$(1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n$$

$$\Rightarrow x(1+x)^n = C_0x + C_1x^2 + C_2x^3 + \dots + C_nx^{n+1} \quad \dots(1)$$

$\Rightarrow$ differentiating w.r.t. 'x' & putting $x = 1$,

$$C_0 + 2C_1 + 3C_2 + \dots + (n+1)C_n = 2^{n-1}(n+2)$$

2. Answer (1)

putting $x = -1$ in last questions equation (1)

$$C_0 - 2C_1 + 3C_2 - \dots + (-1)^n \cdot (n+1)C_n = 0$$

3. Answer (2)

$$y = C_0 + (C_0 + C_1) + (C_0 + C_1 + C_2) + \dots + (C_0 + C_1 + C_2 + \dots + C_{n-1})$$

Adding the terms equidistant from the ends we get

$$y = (C_0 + C_0 + C_1 + C_2 + \dots + C_{n-1}) + (C_0 + C_1 + C_0 + C_1 + \dots + C_{n-2}) + \dots$$

Putting $C_0 = C_n$ in first bracket

Putting $C_0 = C_n$ and $C_1 = C_{n-1}$ in second bracket

And so on

In this way

$$y = (C_0 + C_1 + C_2 \dots C_n) + (C_0 + C_1 + C_2 \dots C_n) + \dots \frac{n}{2} \text{ times}$$

$$y = \frac{n}{2} \cdot 2^n = n \cdot 2^{n-1}$$

Comprehension-III

Let n be a positive integer and

$$(1+x)^n = C_0 + C_1x + C_2x^2 + C_3x^3 + \dots + C_r x^r + \dots + C_{n-1}x^{n-1} + C_n x^n$$

where C_r stands for nC_r then

1. The value of $\sum_{r=0}^n \sum_{s=0}^n (C_r + C_s)$ is

(1) $(n+1)2^{n+1}$

(2) $n \cdot 2^n$

(3) $(n+1)2^n$

(4) $n \cdot 2^{n-1}$

Sol. Answer (1)

$$\sum_{r=0}^n \sum_{s=0}^n (C_r + C_s) = \{(C_0 + C_0) + (C_0 + C_1) + \dots + (C_0 + C_n)\} + \{(C_1 + C_0) + (C_1 + C_1) + (C_1 + C_2) + \dots + (C_1 + C_n)\} + \dots$$

$$+ \{(C_n + C_0) + (C_n + C_1) + \dots + (C_n + C_n)\}$$

$$= 2(n+1)(C_0 + C_1 + C_2 + \dots + C_n)$$

$$= (n+1) \cdot 2 \cdot 2^n = (n+1)2^{n+1}$$

2. The value of $\sum_{0 \leq r < s \leq n} (C_r \pm C_s)^2$ is

(1) $(n \mp 1)^{2n} C_n \pm 2^{2n}$

(2) $(n+1)^{2n} C_n + 2^{2n-1}$

(3) $(n \pm 1)^{2n} C_n \pm 2^{2n}$

(4) $(n+1)^{2n} C_n - 2^{2n-1}$

Sol. Answer (1)

$$\begin{aligned}
 & \sum_{0 \leq r < s \leq n} (C_r \pm C_s)^2 \\
 &= \sum_{0 \leq r < s \leq n} (C_r^2 + C_s^2 \pm 2C_r C_s) \\
 &= \sum_{0 \leq r < s \leq n} (C_r^2 + C_s^2) \pm \sum_{0 \leq r < s \leq n} C_r C_s \\
 &= n(C_0^2 + C_1^2 + \dots + C_n^2) \pm (2^{2n} - 2^n C_n) \\
 &= (C_0 + C_1 + \dots + C_n)^2 = C_0^2 + C_1^2 + \dots + C_n^2 + 2 \sum_{0 \leq r < s \leq n} C_r C_s \\
 &\Rightarrow \sum_{0 \leq r < s \leq n} (C_r \pm C_s)^2 = n \cdot 2^n C_n \pm (2^{2n} - 2^n C_n) \\
 &= (n \mp 1) 2^n C_n \pm 2^{2n}
 \end{aligned}$$

3. The value of $\sum_{r=0}^n \sum_{s=0}^n C_r C_s$ is

(1) 2^{2n}

(2) $n \cdot 2^{2n-1}$

(3) $n \cdot 2^{n+1}$

(4) $n(n-1) \cdot 2^{n-2}$

Sol. Answer (1)

$$\sum_{r=0}^n \sum_{s=0}^n C_r C_s = \left(\sum_{r=0}^n C_r \right) \left(\sum_{s=0}^n C_s \right) = 2^n \cdot 2^n = 2^{2n}$$

Comprehension-IVLet n is a rational number and x is a real number such that $|x| < 1$, then

$$(1+x)^n = 1 + nx + \frac{n(n-1)x^2}{2!} + \frac{n(n-1)(n-2)}{3!} x^3 + \dots$$

This can be used to find the sum of different series.

1. Sum of infinite series

$$1 + \frac{2}{3} \cdot \frac{1}{2} + \frac{2}{3} \cdot \frac{5}{6} \cdot \frac{1}{2^2} + \frac{2}{3} \cdot \frac{5}{6} \cdot \frac{8}{9} \cdot \frac{1}{2^3} + \dots \infty \text{ is}$$

(1) $2^{1/3}$

(2) $4^{1/3}$

(3) $8^{1/3}$

(4) $4^{2/3}$

Sol. Answer (2)

$$\text{We have, } (1+x)^n = 1 + nx + \frac{n(n-1)}{2} x^2 + \dots$$

$$\text{Here, } nx = \frac{2}{3} \cdot \frac{1}{2} \quad \dots (i)$$

$$\frac{n(n-1)}{2}x^2 = \frac{2 \cdot 5}{3 \cdot 6} \cdot \frac{1}{2^2} \quad \dots(ii)$$

Equation (ii) divided by Equation (i)

$$\frac{n(n-1)x^2}{2 \times nx} = \frac{2 \cdot 5 \cdot 1}{3 \cdot 6 \cdot 4} \cdot \frac{3 \cdot 2}{2 \cdot 1}$$

$$\Rightarrow \frac{(n-1)x}{2} = \frac{5}{12}$$

$$\Rightarrow nx - x = \frac{5}{6}$$

$$\Rightarrow x = nx - \frac{5}{6} = \frac{1}{3} - \frac{5}{6} = \frac{2-5}{6} = -\frac{1}{2}$$

$$\therefore n \cdot \frac{-1}{2} = \frac{1}{3} \Rightarrow n = -\frac{2}{3}$$

$$\therefore \text{The require sum} = \left(\frac{1}{2}\right)^{-\frac{2}{3}} = \frac{1}{\left(\frac{1}{2}\right)^{2/3}} = \frac{1}{\left(\frac{1}{4}\right)^{1/3}} = \frac{1}{\frac{1}{4^{1/3}}} = 4^{1/3}$$

2. The sum of the series

$$1 + \frac{1}{3^2} + \frac{1 \cdot 4}{1 \cdot 2} \cdot \frac{1}{3^4} + \frac{1 \cdot 4 \cdot 7}{1 \cdot 2 \cdot 3} \cdot \frac{1}{3^6} + \dots \text{ is}$$

(1) $\sqrt{\frac{3}{2}}$

(2) $\left(\frac{3}{2}\right)^{1/3}$

(3) $\sqrt{\frac{1}{3}}$

(4) $\left(\frac{2}{3}\right)^{1/3}$

Sol. Answer (2)

In this problem

$$nx = \frac{1}{3^2}$$

$$\frac{n(n-1)}{2}x^2 = \frac{1 \cdot 4}{1 \cdot 2} \cdot \frac{1}{3^4}$$

Solving above two we get,

$$n = -\frac{1}{3}, \text{ and } x = -\frac{1}{3}$$

$$\therefore \text{Required sum} = \left(1 - \frac{1}{3}\right)^{-\frac{1}{3}} = \left(\frac{2}{3}\right)^{-\frac{1}{3}} = \frac{1}{\left(\frac{2}{3}\right)^{1/3}} = \left(\frac{3}{2}\right)^{1/3}$$

SECTION - D

Assertion-Reason Type Questions

1. STATEMENT-1 : The number of distinct term in the expansion of $(1 + px)^{20} + (1 - px)^{20}$ is 42.

and

STATEMENT-2 : Number of term in the expansion of $(1 + x)^n$ is $(n + 1)$.

Sol. Answer (4)

$$(1 + px)^{20} = {}^{20}C_0 + {}^{20}C_1(px) + {}^{20}C_2(px)^2 + \dots + {}^{20}C_{20}(px)^{20} \quad \dots(i)$$

$$(1 - px)^{20} = {}^{20}C_0 - {}^{20}C_1(px) + {}^{20}C_2(px)^2 - \dots + {}^{20}C_{20}(px)^{20} \quad \dots(ii)$$

Adding the equations

$$(1 + px)^{20} + (1 - px)^{20} = 2[{}^{20}C_0 + {}^{20}C_2(px)^2 + {}^{20}C_4(px)^4 + \dots + {}^{20}C_{20}(px)^{20}]$$

Hence total terms is 11.

Hence statement-1 is false but statement-2 is true.

2. STATEMENT-1 : The coefficient of $a^3b^4c^3$ in the expansion of $(a - b + c)^{10}$ is $\frac{10!}{3!4!3!}$.

and

STATEMENT-2 : The coefficient of $x^p y^q z^r$ in the expansion of $(x + y + z)^n$ is $\frac{n!}{p!q!r!}$ for all integer n .

Sol. Answer (3)

Statement-1 is correct because coefficient is

$$(-1)^4 \frac{10!}{3!4!3!} = \frac{10!}{3!4!3!}$$

But statement-2 is false because no conditions is given on p, q, r

If $p + q + r \neq n$, then coefficient will be zero.

3. STATEMENT-1 : If $\sum_{r=1}^n r^3 \left(\frac{{}^nC_r}{{}^nC_{r-1}} \right)^2 = 196$, then the sum of the coefficients of power of x in the expansion of

the polynomial $(x - 3x^2 + x^3)^n$ is -1 .

and

STATEMENT-2 : $\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r} \quad \forall n \in N \text{ and } r \in W$.

Sol. Answer (4)

$$\sum_{r=1}^n r^3 \left(\frac{n-r+1}{r} \right)^2 = \sum_{r=1}^n r(n-r+1)^2$$

$$= \sum_{r=1}^n r \{ (n+1)^2 - 2(n+1)r + r^2 \}$$

$$\begin{aligned}
 &= (n+1)^2 \sum_{r=1}^n r - 2(n+1) \sum_{r=1}^n r^2 + \sum_{r=1}^n r^3 \\
 &= (n+1)^2 \cdot \frac{n(n+1)}{2} - \frac{2(n+1) \cdot n(n+1)(2n+1)}{6} + \left(\frac{n(n+1)}{2} \right)^2 \\
 &\Rightarrow \frac{(n+1)^2 \cdot n(n+2)}{12} = 14^2 \\
 &\Rightarrow \boxed{n=6}
 \end{aligned}$$

$\therefore$ Sum of coefficients = $(1 - 3 + 1)^6 = 1$

4. STATEMENT-1 : The number of terms in the expansion of $\left(x + \frac{1}{x} + 1\right)^n$ is $2n + 1$.

and

STATEMENT-2 : The number of terms in the expansion of $(a_1 + a_2 + \dots + a_m)^n$ is $n + m - 1 C_{m-1}$.

Sol. Answer (2)

Statement-1:

$$\begin{aligned}
 \text{Given expression is } &= \left(1 + \left(x + \frac{1}{x}\right)\right)^n \\
 &= {}^nC_0 \cdot 1^n \cdot \left(x + \frac{1}{x}\right)^0 + {}^nC_1 \cdot 1^{n-1} \left(x + \frac{1}{x}\right)^1 + {}^nC_2 \cdot 1^{n-2} \left(x + \frac{1}{x}\right)^2 + \dots + \left(x + \frac{1}{x}\right)^n
 \end{aligned}$$

This will be of the form $= a_0 + a_1x + a_2x^2 + \dots + a_nx^n + \frac{b_1}{x} + \frac{b_2}{x^2} + \dots + \frac{b_n}{x^n}$

Hence, number of term = $n + n + 1 = 2n + 1$

Statement-2, obviously the number of terms in the expansion of

$(a_1 + a_2 + \dots + a_n)^n$ is $n + m - 1 C_{m-1}$

Hence Statement-1 and Statement-2 both are true but (2) is not the correct explanation of (1).

5. STATEMENT-1 : Sum of the coefficients of last 30 terms in the expansion of $(1 + x)^{49}$, when expanded in ascending powers of x , is 2^{48} .

and

STATEMENT-2 : P^{th} term from the end in the expansion of $(x + y)^n$ is $(n - P + 2)^{\text{th}}$ term from the beginning.

Sol. Answer (4)

The sum of last 25 terms in the expansion of $(1 + x)^{49}$ will be 2^{48} . Hence, sum of 30 terms will be obviously greater than 2^{48} .

Hence, Statement-1 is false but Statement-2 is true.

6. STATEMENT-1 : In the expansion of $(\sqrt{5} + 3^{1/5})^{10}$, sum of integral terms is 3134.

and

$$\text{STATEMENT-2 : } (x + y)^n = \sum_{r=0}^n {}^nC_r \cdot x^{n-r} y^r.$$

Sol. Answer (1)

Sum of integral term

$$= {}^{10}C_0 \cdot (5^{1/2})^0 (3^{1/5})^{10} + {}^{10}C_{10} \cdot (5^{1/2})^{10} \cdot (3^{1/5})^0$$

$$= 3^2 + 5^5$$

$$= 9 + 3125 = 3134$$

Hence, Statement-1 is true and Statement-2 is also true and Statement-2 is the correct explanation of Statement-1.

SECTION - E

Matrix-Match Type Questions

Let n be a positive integer and $(1 + x)^n = C_0 + C_1x + C_2x^2 + C_3x^3 + \dots + C_nx^n$.

1. Match the following:

Column-I

Column-II

(A) If the second term of the expansion $(x^{\frac{1}{13}} + \frac{x}{\sqrt{x-1}})^n$ (p) 5

is $14x^{\frac{5}{2}}$, then the value of $\frac{{}^nC_3}{{}^nC_2}$ is

(B) In the binomial $(2^{\frac{1}{3}} + 3^{-\frac{1}{3}})^n$, if the ratio of the (q) 7

seventh term from beginning of the expansion to the seventh term from its end is $1/6$, then n is equal to

(C) Given that the term of the expansion $(x^{\frac{1}{3}} - x^{-\frac{1}{2}})^{15}$ (r) 4

which does not contain x is $5m$, where $m \in N$, then

$\frac{m}{143}$ is equal to

(D) If the coefficients of x^7 and x^8 in the expansion of (s) 9

$(2 + \frac{x}{3})^n$ are equal, then the value of $\frac{n}{11}$ is

Sol. Answer A(r), B(s), C(q), D(p)

$$(A) T_2 = {}^nC_1 \left(x^{\frac{1}{13}} \right)^{n-1} \left(\frac{x}{x^{\frac{1}{2}}} \right)^1 = 14x^{\frac{5}{2}}$$

$$\Rightarrow {}^nC_1 x^{\frac{n-1}{13} + \frac{3}{2}} = 14x^{\frac{5}{2}}$$

$$\Rightarrow n = 14$$

$$\text{Hence, } \frac{{}^nC_3}{{}^nC_2} = \frac{{}^{14}C_3}{{}^{14}C_2} = \frac{14-3+1}{3} = 4$$

$$(B) \frac{T_7 \text{ from the beginning}}{T_7 \text{ from the end}} = \frac{{}^nC_6 \left(\frac{1}{2^3} \right)^{n-6} \left(3^{-\frac{1}{3}} \right)^6}{{}^nC_{n-6} \left(-3^{-\frac{1}{3}} \right)^{n-6} \left(\frac{1}{2^3} \right)^6} = \frac{1}{6}$$

$$\Rightarrow \left(\frac{1}{2^3} \right)^{n-6-6} \left(3^{-\frac{1}{3}} \right)^{6-n+6} = \frac{1}{6}$$

$$\Rightarrow 2^{\frac{n-12}{3}} \cdot 3^{\frac{12-n}{6}} = 2^{-1} 3^{-1}$$

$$\Rightarrow n - 12 = -3 \text{ or } \frac{12-n}{6} = 1$$

$$\Rightarrow n = 9$$

$$(C) T_{r+1} = {}^{15}C_r \left(x^{\frac{1}{3}} \right)^{15-r} \left(-x^{-\frac{1}{2}} \right)^r$$

$$= {}^{15}C_r x^{\frac{15-r}{3}} (-1)^{\frac{r}{2}} x^{-\frac{r}{2}}$$

$$\Rightarrow \frac{15-r}{3} - \frac{r}{2} = 0$$

$$\Rightarrow 30 - 2r - 3r = 0 \Rightarrow r = 6$$

$$\text{Coefficient of term} = {}^{15}C_6 = 5m$$

$$\Rightarrow \frac{15 \times 14 \times 13 \times 12 \times 11}{6 \times 5 \times 4 \times 3 \times 2 \times 1} = 5$$

$$\Rightarrow m = 7 \times 143$$

$$\frac{m}{143} = 7$$

(D) Coefficient of x^7 = Coefficient of x^8

$${}^nC_7 2^{n-7} \left(\frac{1}{3}\right)^7 = {}^nC_8 2^{n-8} \left(\frac{1}{3}\right)^8$$

$$\Rightarrow 2 \cdot {}^nC_7 = {}^nC_8 \cdot \left(\frac{1}{3}\right)$$

$$\Rightarrow 6 \cdot {}^nC_7 = {}^nC_8$$

$$\frac{n-8+1}{8} = 6 \Rightarrow \frac{n-7}{8} = 6 \Rightarrow n = 55$$

$$\frac{n}{11} = 5$$

2. Match the following:

Column-I

- (A) The remainder when 2^{2007} is divided by 17 is
- (B) The greatest value of the term independent of x in the expansion of $(x \sin p + x^{-1} \cos p)^{10}$, $p \in R$ is equal

to k , then $\frac{8k}{9}$ is equal to

- (C) In the expansion of $(7^{\frac{1}{3}} + 11^{\frac{1}{9}})^{2007}$, the number of

rational terms is r then $\frac{r}{14}$ is equal to

- (D) If $(1 + x + x^2)^n = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots + a_{2n}x^{2n}$ then $8[a_0 - a_2 + a_4 - a_6 + \dots]^2$, if $n = 4p + 2$ is equal to

Column-II

(p) 8

(q) 9

(r) 7

(s) 16

Sol. Answer A(q), B(r), C(s), D(p)

$$\begin{aligned} \text{(A)} \quad y &= 2^{2007} = 2^3(2^{2004}) = 2^3(16)^{501} \\ &= 2^3(17-1)^{501} = -2^3(1-17)^{501} \\ &= -8(1 - {}^{501}C_1(17) + {}^{501}C_2(-17)^2 + \dots + {}^{501}C_{501}(-17)^{501}) \\ &= -8 + 17 \times I, \text{ where } I \text{ is an integer.} \end{aligned}$$

$\Rightarrow$ When p is divided by 17, then remainder = $17 - 8 = 9$

$$\begin{aligned} \text{(B)} \quad T_{r+1} &= {}^{10}C_r (x \sin p)^{10-r} (x^{-1} \cos p)^r \\ &= {}^{10}C_r x^{10-r} x^{-r} (\sin p)^{10-r} (\cos p)^r \end{aligned}$$

For independent term

$$10 - r - r = 0 \Rightarrow r = 5$$

$\Rightarrow$ The coefficient of independent term

$$= (\sin p)^5 (\cos p)^5 \times {}^{10}C_5$$

$$= \frac{1}{32} (2 \sin p \cos p)^5 \times {}^{10}C_5$$

$$= \frac{1}{32} (\sin 2p)^5 \times {}^{10}C_5$$

Maximum value of coefficient

$$= \frac{1}{32} \times 1 \times \frac{10 \times 9 \times 8 \times 7 \times 6}{5 \times 4 \times 3 \times 2 \times 1}$$

$$K = \frac{63}{8}$$

$$\Rightarrow \frac{8K}{9} = \frac{8}{9} \times \frac{63}{8} = 7$$

$$\text{(C)} \quad \left(7^{\frac{1}{3}} + 11^{\frac{1}{9}} \right)^{2007}$$

Let general term is T_{r+1}

$$T_{r+1} = {}^{2007}C_r \left(7^{\frac{1}{3}} \right)^{2007-r} (11)^{\frac{r}{9}} = {}^{2007}C_r 7^{\frac{2007-r}{3}} (11)^{\frac{r}{9}}$$

For rational term both $\frac{2007-r}{3}$ and $\frac{r}{9}$ must be integer

If $\frac{r}{9}$ is integer then $r = 9, 18, 27, 36, \dots, 2007$

At these values of r $\frac{2007-r}{3}$ is integer

Hence total term are $\frac{2007}{9} + 1 = 224$

$$r = 224$$

$$\frac{r}{14} = \frac{224}{14} = 16$$

$$(D) (1 + x + x^2)^n = a_0 + a_1x + a_2x^2 + \dots a_{2n}x^{2n}$$

$$\text{Putting } x = i = \sqrt{-1}$$

$$(1 + i + i^2)^n = a_0 + a_1i + a_2i^2 + a_3i^3 + \dots a_{2n}(i)^{2n}$$

$$\Rightarrow (i)^{2n} = (a_0 - a_2 + a_4 - a_6 + \dots) + i(a_1 - a_3 + a_5 - a_7 + \dots)$$

$$n = 4p + 2$$

$$\Rightarrow (i)^{8p+4} = (a_1 - a_2 + a_4 - a_6 + \dots) + i(a_1 - a_3 + a_5 - a_7 + \dots)$$

$$\text{Comparing } a_0 - a_2 + a_4 - a_6 + \dots = 1$$

$$8[a_0 - a_2 + a_4 - a_6 + \dots] = 8$$

3. Match the following:

Column-I

$$(A) (3 + x^{10} + x^{11})^{2009} = a_0 + a_1x + a_2x^2 + a_3x^3 \dots,$$

$$\text{then } a_0 - \frac{a_1}{2} - \frac{a_2}{2} + a_3 - \frac{a_4}{2} - \frac{a_5}{2} + a_6 \dots$$

will be equal to

$$(B) (3 + x^2 + x^3)^{2008} = a_0 + a_1x + a_2x^2 + \dots, \text{ then } a_0 + a_1 \text{ is equal to}$$

$$(C) (3 - x + x^2)^{2009} = a_0 + a_1x + a_2x^2 + \dots, \text{ then } a_0 - a_1 \text{ is equal to}$$

$$(D) \text{ In expansion of } \left(x + \frac{1}{x}\right)^{2009} \text{ the coefficient of middle}$$

term may be

Column-II

$$(p) 2^{2009}$$

$$(q) 3^{2008}$$

$$(r) {}^{2009}C_{1004}$$

$$(s) {}^{2009}C_{1005}$$

$$(t) 2 \cdot 3^{2008}$$

Sol. Answer A(p), B(q), C(t), D(r, s)

$$(A) (3 + x^{10} + x^{11})^{2009} = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots$$

$$\text{Put } x = \omega \text{ and } \omega^2$$

$$(3 + \omega + \omega^2)^{2009} = a_0 + a_1\omega + a_2\omega^2 + a_3\omega^3 + \dots$$

$$(3 + \omega^2 + \omega)^{2009} = a_0 + a_1\omega^2 + a_2\omega + a_3 + a_4\omega^2 + \dots = 2^{2009}$$

On adding

$$2^{2009} + 2^{2009} = 2a_0 - a_1 - a_2 + 2a_3 - a_4 - a_5 + 2a_6 + \dots = 2^{2009}$$

$$\Rightarrow a_0 - \frac{a_1}{2} - \frac{a_2}{2} + a_3 - \frac{a_4}{2} - \frac{a_5}{2} + 96 \dots = 2^{2009}$$

$$(B) (3 + x^2 + x^3)^{2008} = a_0 + a_1x + a_2x^2 + \dots$$

$$\text{Put } x = 0$$

$$a_0 = 3^{2008}, \text{ on differentiating and putting } x = 1$$

$$2008(3 + x^2 + x^3)^{2007} (2x + 3x^2) = a_1 + 2a_2x + \dots$$

$$\text{Put } x = 0 \Rightarrow a_1 = 0$$

$$a_0 + a_1 = 3^{2008}$$

$$(C) (3 - x + x^2)^{2009} = a_0 + a_1x + a_2x^2 + \dots$$

$$\text{Put } x = 0 \Rightarrow a_0 = 3^{2009}$$

On differentiating

$$2009 (3 - x + x^2)^{2008} (-1 + 2x) = a_1 + 2a_2x + \dots$$

$$\text{Put } x = 0$$

$$a_1 = 2009 (3)^{2008} (-1)$$

$$a_1 = -2009 (3)^{2008}$$

$$a_0 - a_1 = 3^{2009} + 2009 \cdot 3^{2008}$$

$$= 3^{2008} (3 + 2009)$$

$$= 2012 \cdot 3^{2008}$$

$$(D) \text{ Coefficient of middle term} = {}^{2009}C_{1005} = {}^{2009}C_{1004}$$

4. Match the entries of Column-I with those of Column-II.

Column-I

Column-II

(A) If the fourth term in the expression of $\left[\sqrt{x^{\left(\frac{1}{\log_{10} x+1}\right)} + x^{\frac{1}{12}}} \right]^6$ equal (p) 1

to 200 and $x > 1$, then x equals

(B) The number of non-zero terms in the expansion of (q) 5

$$\left[(1+3\sqrt{2}x)^9 + (1-3\sqrt{2}x)^9 \right] \text{ is}$$

(C) If $\frac{T_2}{T_3}$ in the expansion of $(a+b)^n$ and $\frac{T_3}{T_4}$ in the expansion of (r) 10

$(a+b)^{n+3}$ are equal, then n is equal to

(D) The digit of unit's place in the number $17^{1995} + 11^{1995} - 7^{1995}$ is (s) 5C_2

(t) 5C_1

Sol. Answer A(r, s), B(q, t), C(q, t), D(p)

(A) $T_4 = 200$

$$\Rightarrow {}^6C_3 \left[x^{\frac{3}{2(\log x+1)}} \cdot x^{\frac{1}{4}} \right] = 200$$

$$\Rightarrow \frac{3}{x^{2(\log x + 1)}} + \frac{1}{4} = 10$$

$$\Rightarrow \frac{3}{2(\log x + 1)} + \frac{1}{4} = \frac{1}{\log_{10} x}$$

$$\Rightarrow \frac{6 + \log x + 1}{4(\log x + 1)} = \frac{1}{\log_{10} x}$$

Now let $\log_{10} x = y$

$$\Rightarrow \frac{7 + y}{4(y + 1)} = \frac{1}{y}$$

$$\Rightarrow 4y + y^2 = 4y + 4$$

$$\Rightarrow y^2 + 3y - 4 = 0$$

$$\Rightarrow y^2 + 4y - y - 4 = 0$$

$$\Rightarrow y = 1, -4$$

$$\Rightarrow x = 10^1, 10^{-4}$$

$$\because x > 1$$

$$\therefore x = 10$$

(B) Required expansion = $2[T_1 + T_3 + T_5 + T_7 + T_9]$

$$\therefore \text{Number of non-zero terms} = 5$$

(C) Since $\frac{T_{r+1}}{T_r} = \frac{n-r+1}{r} \left(\frac{b}{a}\right)$

$$\therefore \frac{T_3}{T_2} = \frac{n-1}{2} \left(\frac{b}{a}\right)$$

$$\frac{T_4}{T_3} = \frac{n+1}{3} \left(\frac{b}{a}\right)$$

Now,

$$\frac{n-1}{2} = \frac{n+1}{3}$$

$$\Rightarrow 3n - 3 = 2n + 2$$

$$\Rightarrow n = 5$$

(D) $17^{1995} + 11^{1995} - 7^{1995}$

$$= (7 + 10)^{1995} + (10 + 1)^{1995} - (7)^{1995}$$

$$= ({}^{1995}C_1 \cdot 7^{1994} \cdot 10^1 + \dots + 10^{1995}) + ({}^{1995}C_1 \cdot 10^1 + \dots + {}^{1995}C_{1995} \cdot 10^{1995}) + 1$$

$$= (\text{a multiple of } 10) + 1$$

Hence the units place digit = 1.

SECTION - F

Integer Answer Type Questions

1. If the coefficient of x^3 is 1140 in the expansion of $(1 + 2x + kx^2)^{10}$, then the value of k is _____.

Sol. Answer (1)

General term in the expansion of

$$(1 + 2x + kx^2)^{10} \text{ is } = \frac{10!}{p!q!r!} \cdot 1^p \cdot (2x)^q \cdot (kx^2)^r = \frac{10!}{p!q!r!} \cdot 2^q \cdot k^r \cdot x^{q+2r}$$

Under the condition

$$p + q + r = 10$$

$$q + 2r = 3$$

p	q	r
7	3	0
8	1	1

Now, the coefficient of x^3 in $(1 + 2x + kx^2)^{10}$

$$= \frac{10!}{7!3!0!} \cdot 2^3 \cdot k^0 + \frac{10!}{8!1!1!} \cdot 2^1 \cdot k^1$$

$$= 960 + 180k$$

$$\text{Now, } 960 + 180k = 1140$$

$$\Rightarrow 180k = 180$$

$$\Rightarrow k = 1$$

2. The sum $\sum_{i=0}^{3m} \binom{10}{i} \binom{20}{3m-i}$, where $\binom{p}{q} = 0$ if $p < q$ is maximum then m is _____.

Sol. Answer (5)

$$\sum_{i=0}^{3m} \binom{10}{i} \binom{20}{3m-i} = \sum_{i=0}^{3m} {}^{10}C_i \cdot {}^{20}C_{3m-i}$$

$$= \text{Coefficient of } x^{3m} \text{ in the expansion of } (1+x)^{10}(1+x)^{20} = {}^{30}C_{3m}$$

$$\text{It is maximum when } 3m = \frac{30}{2}$$

$$\Rightarrow m = \frac{15}{3} = 5$$

3. The coefficients of three consecutive terms of $(1+x)^{n+5}$ are in the ratio 5 : 10 : 14. Then $n =$ _____.

[JEE(Advanced) 2013]

Sol. Answer (6)

Let consecutive terms be t_{r+2}, t_{r+1}, t_r

$$\text{So, } \frac{t_{r+1}}{t_r} = \frac{10}{5}$$

$$\Rightarrow \frac{(n+5)-(r+1)+1}{r+1} = 2$$

$$\Rightarrow n - 3r + 3 = 0 \quad \dots(i)$$

$$\text{also, } \frac{t_{r+2}}{t_{r+1}} = \frac{14}{10}$$

$$\Rightarrow 5n - 12r + 6 = 0 \quad \dots(ii)$$

Solving, $n = 6$

4. The coefficient of x^9 in the expansion of $(1+x)(1+x^2)(1+x^3)\dots(1+x^{100})$ is _____

[JEE(Advanced) 2015]

Sol. Answer (8)

$$(1+x)(1+x^2)\dots(1+x^{100})$$

Possible combination of 9 are (0, 9), (1, 8), (2, 7), (3, 6), (4, 5), (1, 2, 6), (1, 3, 5) (2, 3, 4)

So, coefficient = 8

SECTION - G

Multiple True-False Type Questions

1. STATEMENT-1 : The integral part of $(8+3\sqrt{7})^{20}$ is even.

STATEMENT-2 : The sum of the last eight coefficients in the expansion of $(1+x)^{16}$ is 2^{15} .STATEMENT-3 : If $R = (5\sqrt{5}+11)^{2n+1} = [R] + F$, where $[R]$ denotes the greatest integer in R , then $RF = 2^{2n+1}$.

(1) F F F

(2) F F T

(3) T F F

(4) T F T

Sol. Answer (1)

STATEMENT-1 : $(8+3\sqrt{7})^{20} = I + F$ where I is an integer and F is fraction.Now let, $G = (8-3\sqrt{7})^{20}$, then $0 < G < 1$

$$\text{Now, } I + F + G = (8+3\sqrt{7})^{20} + (8-3\sqrt{7})^{20}$$

$$= 2 \left[{}^{20}C_0 \cdot 8^{20} \cdot (3\sqrt{7})^0 + {}^{20}C_2 \cdot 8^{18} \cdot (3\sqrt{7})^2 + \dots + {}^{20}C_{20} \cdot 8^0 \cdot (3\sqrt{7})^{20} \right]$$

= An even integer

$$\Rightarrow F + G = 1$$

$$\Rightarrow I + 1 = \text{An even integer}$$

$\Rightarrow l = \text{An odd integer.}$

Hence, Statement-1 is false.

STATEMENT-2 : We have

$${}^{16}C_0 + {}^{16}C_1 + \dots + {}^{16}C_{16} = 2^{16}$$

$$\text{Now let } {}^{16}C_1 + \dots + {}^{16}C_7 = {}^{16}C_9 + \dots + {}^{16}C_{16} = x$$

$$\Rightarrow x + {}^{16}C_8 + x = 2^{16}$$

$$\Rightarrow x = 2^{15} - \frac{{}^{16}C_8}{2}$$

Hence false.

STATEMENT-3 : Let $G = (5\sqrt{5} - 11)^{2n+1}$

$$[R] + F - G = (5\sqrt{5} + 11)^{2n+1} - (5\sqrt{5} - 11)^{2n+1}$$

= An even integer

$$\Rightarrow [R] + F - G = \text{An even integer}$$

$$\therefore \boxed{F = G}$$

$$\therefore RF = RG = 4^{2n+1}$$

Hence, Statement-3 is false

2. STATEMENT-1 : The middle term of $\left(x + \frac{1}{x}\right)^{2n}$ can exceed $\frac{(2n)^n}{n!}$ for some value of x .

STATEMENT-2 : The coefficient of x^n in the expansion of $(1 - 2x + 3x^2 - 4x^3 + \dots)^{-n}$ is $\frac{1 \cdot 3 \cdot 5 \dots (2n-1)}{n!} \cdot 2^n$.

STATEMENT-3 : The coefficient of x^5 in $(1 + 2x + 3x^2 + \dots)^{-3/2}$ is 21.

(1) T T F

(2) F T T

(3) F T F

(4) F F F

Sol. Answer (3)

STATEMENT-1 : There will be $2n + 1$ terms in the expansion of $\left(x + \frac{1}{x}\right)^{2n}$

Hence middle term is $= {}^{2n}C_n$

$$= \frac{2n!}{n!n!}$$

$$= \frac{2n!}{n!n!} = \frac{1 \cdot 3 \cdot 5 \dots (2n-1)}{n!} \cdot 2^n$$

But A.M. $\geq$ G.M.

$$\frac{1+3+5+\dots+(2n-1)}{n} \geq (1 \cdot 3 \cdot 5 \dots (2n-1))^{\frac{1}{n}}$$

$$\Rightarrow \frac{n^2}{n} \geq (1 \cdot 3 \dots (2n-1))^{\frac{1}{n}}$$

$$\Rightarrow n^n \geq 1 \cdot 3 \cdot 5 \dots (2n-1)$$

$$\Rightarrow n^n \cdot 2^n \geq 1 \cdot 3 \cdot 5 \dots (2n-1) 2^n$$

$$\Rightarrow \frac{(2n)^n}{n!} \geq \text{middle term}$$

Hence Statement-1 is false.

STATEMENT-2 : We have, $(1 - 2x + 3x^2 - 4x^3 + \dots)^{-n} = [(1+x)^{-2}]^{-n} = (1+x)^{2n}$

Coefficient of x^n in $(1+x)^{2n}$ is

$$= {}^{2n}C_n = \frac{2n!}{n!n!}$$

$$= \frac{1 \cdot 3 \cdot 5 \dots (2n-1)}{n!} \cdot 2^n$$

Hence it is true.

STATEMENT-3 : We have, $(1+2x+3x^2+\dots)^{-\frac{3}{2}} = [(1-x)^{-2}]^{-\frac{3}{2}} = (1-x)^3$

So, coefficient of $x^5 = 0$

Hence Statement-3 is false.

SECTION - H

Aakash Challengers Questions

1. If $(1+x)^{15} = C_0 + C_1x + C_2x^2 + C_3x^3 + \dots + C_{15}x^{15}$ and $(k = C_2 + 2C_3 + 3C_4 + \dots + 14C_{15})$ then the value of $\frac{k-993}{1000}$ is equal to

Sol. $(1+x)^{15} = C_0 + C_1x + C_2x^2 + \dots + C_{15}x^{15}$

$$\frac{(1+x)^{15}}{x} = \frac{C_0}{x} + C_1 + C_2x + C_3x^2 + \dots + C_{15}x^{14}$$

Differentiating with respect to x

$$\frac{(1+x)^{15} - 15(1+x)^{14}x}{x^2} = -\frac{C_0}{x^2} + 0 + C_2x + 2C_3x + 3C_4x^2 + \dots + 14C_{15}x^{13}$$

Putting $x = 1$, we get

$$\frac{2^{15} - 15 \cdot 2^{14}}{1} = -1 + k$$

$$\Rightarrow k = 2^{15} - 15 \cdot 2^{14} + 1$$

$$\Rightarrow k - 993 = 2^{15} - 15 \cdot 2^{14} - 992$$

$$\Rightarrow \frac{k-993}{1000} = \frac{2^{15} - 15 \cdot 2^{14} - 992}{1000} = 212$$

2. If C_r stands for nC_r , then the sum of the first $(n + 1)$ terms of the series $aC_0 - (a + d)C_1 + (a + 2d)C_2 - (a + 3d)C_3 + \dots$, is equal to

Sol. $aC_0 - (a + d)C_1 + (a + 2d)C_2 - (a + 3d)C_3 + \dots$

$$= a(C_0 - C_1 + C_2 - C_3 + \dots) - d(C_1 - 2C_2 + 3C_3 - 4C_4 + \dots)$$

$$= a \times 0 - d \times 0 = 0$$

3. In the expansion of $(1 + x)^{10} = {}^{10}C_0 + {}^{10}C_1x + {}^{10}C_2x^2 + \dots + {}^{10}C_{10}x^{10}$, the value of

$$528 \left[\frac{{}^{10}C_0}{2} - \frac{{}^{10}C_1}{3} + \frac{{}^{10}C_2}{4} - \frac{{}^{10}C_3}{5} + \dots + \frac{{}^{10}C_{10}}{12} \right] \text{ is equal to$$

Sol. $(1 + x)^{10} = {}^{10}C_0 + {}^{10}C_1x + {}^{10}C_2x^2 + \dots + {}^{10}C_{10}x^{10}$

$$x(1 + x)^{10} = {}^{10}C_0x + {}^{10}C_1x^2 + {}^{10}C_2x^3 + \dots + {}^{10}C_{10}x^{11}$$

$$\int x(1 + x)^{10} dx = \int ({}^{10}C_0x + {}^{10}C_1x^2 + {}^{10}C_2x^3 + \dots) dx$$

$$\int x(1 + x)^{10} dx = \int ({}^{10}C_0x + {}^{10}C_1x^2 + \dots + {}^{10}C_{10}x^{11}) dx$$

$$\int x(1 + x)^{10} dx = {}^{10}C_0 \frac{x^2}{2} + {}^{10}C_1 \frac{x^3}{3} + {}^{10}C_2 \frac{x^4}{4} + \dots + {}^{10}C_{10} \frac{x^{12}}{12} + k$$

$$= \frac{x(1 + x)^{11}}{11} - \int \frac{(1 + x)^{11}}{11} dx = {}^{10}C_0 \frac{x^2}{2} + \frac{{}^{10}C_1x^3}{3} + \dots + \frac{{}^{10}C_{10}x^{12}}{12} + k$$

$$\Rightarrow \frac{x(1 + x)^{11}}{11} - \frac{(1 + x)^{12}}{132} = {}^{10}C_0 \frac{x^2}{2} + \frac{{}^{10}C_1x^3}{3} + \dots + \frac{{}^{10}C_{10}x^{12}}{12} + k$$

Put $x = 0$

$$-\frac{1}{132} = k$$

$$\Rightarrow \frac{x(1 + x)^{11}}{11} - \frac{(1 + x)^{12}}{132} = {}^{10}C_0 \frac{x^2}{2} + \frac{{}^{10}C_1x^3}{3} + \dots + \frac{{}^{10}C_{10}x^{12}}{12} - \frac{1}{132}$$

At $x = -1$

$$\Rightarrow 0 = \frac{{}^{10}C_0}{2} - \frac{{}^{10}C_1}{3} + \frac{{}^{10}C_2}{4} - \frac{{}^{10}C_3}{5} + \dots + \frac{{}^{10}C_{10}}{12} - \frac{1}{132}$$

$$\Rightarrow \frac{{}^{10}C_0}{2} - \frac{{}^{10}C_1}{3} + \frac{{}^{10}C_2}{4} - \frac{{}^{10}C_3}{5} + \dots = \frac{1}{132}$$

$$528 \left[\frac{{}^{10}C_0}{2} - \frac{{}^{10}C_1}{3} + \frac{{}^{10}C_2}{4} - \frac{{}^{10}C_3}{5} + \dots \right] = 528 \times \frac{1}{132} = 4$$

4. Let $\sum_{r=0}^{2010} a_r x^r = (1 + x + x^2 + x^3 + x^4 + x^5)^{402}$ and $\sum_{r=0}^{2010} a_r = a$, then the value of $\left(\frac{\sum_{r=0}^{2010} r \cdot a_r}{\sum_{r=0}^{2010} a_r} \right)$ is equal to

Sol. $(1 + x + x^2 + x^3 + x^4 + x^5)^{402} = \sum_{r=0}^{2010} a_r x^r \quad \dots(i)$

At $x = 1$, $6^{402} = \sum_{r=0}^{2010} a_r \quad \dots(ii)$

Differentiating equation (i) with respect to x

$$\Rightarrow 402(1 + x + x^2 + x^3 + x^4 + x^5)^{401}(1 + 2x + 3x^2 + 4x^3 + 5x^4)$$

$$= \sum_{r=0}^{2010} r \cdot a_r \cdot x^{r-1}$$

At $x = 1$, we get

$$402 \times 6^{401} \left(\frac{5 \times 6}{2} \right) = \sum_{r=0}^{2010} r \cdot a_r \quad \dots (iii)$$

Dividing equation (iii) by (ii)

$$\frac{\sum_{r=0}^{2010} r \cdot a_r}{\sum_{r=0}^{2010} a_r} = \frac{(402) \times 6^{401} \times 15}{6^{402}}$$

$$= \frac{402}{6} \times 15$$

$$= 67 \times 15$$

$$= 1005$$

5. Find ${}^nC_1 - \frac{1}{2} {}^nC_2 + \frac{1}{3} {}^nC_3 - \dots + (-1)^{n-1} \frac{1}{n} {}^nC_n$.

Sol. Let $S_n = {}^nC_1 - \frac{1}{2} {}^nC_2 + \frac{1}{3} {}^nC_3 - \dots + (-1)^{n-1} \frac{1}{n} {}^nC_n$

$$S_{n-1} = {}^{n-1}C_1 - \frac{1}{2} {}^{n-1}C_2 + \frac{1}{3} {}^{n-1}C_3 - \dots + (-1)^{n-2} \frac{1}{n-1} {}^{n-1}C_{n-1}$$

$$\therefore S_n - S_{n-1} = {}^nC_1 - {}^{n-1}C_1 - \frac{1}{2} ({}^nC_2 - {}^{n-1}C_2) + \frac{1}{3} ({}^nC_3 - {}^{n-1}C_3) + \dots$$

$$\dots + (-1)^{n-2} \cdot \frac{1}{n-1} ({}^nC_{n-1} - {}^{n-1}C_{n-1}) + (-1)^{n-1} \cdot \frac{1}{n} {}^nC_n = \frac{1}{n}$$

Thus, we get $S_n - S_{n-1} = \frac{1}{n}$

Putting $n = 2, 3, \dots, n$ we get

$$S_2 - S_1 = \frac{1}{2}$$

$$S_3 - S_2 = \frac{1}{3}$$

$$S_4 - S_3 = \frac{1}{4}$$

⋮

$$S_n - S_{n-1} = \frac{1}{n}$$

$$S_n - S_1 = \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{n}$$

$$\Rightarrow S_n = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{n}$$

6. Prove that $\sum_{r=1}^{2n-1} (-1)^{r-1} \cdot \frac{r}{{}^{2n}C_r} = \frac{n}{n+1}$

Sol. L.H.S. = $\sum_{r=1}^{2n-1} (-1)^{r-1} \cdot \frac{r}{{}^{2n}C_r}$

$$= \frac{1}{{}^{2n}C_1} - \frac{2}{{}^{2n}C_2} + \frac{3}{{}^{2n}C_3} - \frac{4}{{}^{2n}C_4} + \dots + \frac{2n-1}{{}^{2n}C_{2n-1}}$$

$$= \left(\frac{1}{{}^{2n}C_1} + \frac{2n-1}{{}^{2n}C_{2n-1}} \right) - \left(\frac{2}{{}^{2n}C_2} + \frac{2n-2}{{}^{2n}C_{2n-2}} \right) + \left(\frac{3}{{}^{2n}C_3} + \frac{2n-3}{{}^{2n}C_{2n-3}} \right) - \dots + (-1)^{n-1} \frac{n}{{}^{2n}C_n}$$

$$= \left\{ \frac{2n}{{}^{2n}C_1} - \frac{2n}{{}^{2n}C_2} + \frac{2n}{{}^{2n}C_3} - \dots \text{to } n-1 \text{ terms} \right\} + (-1)^{n-1} \cdot \frac{n}{{}^{2n}C_n}$$

$$= 2n \left[\frac{1}{{}^{2n}C_1} - \frac{1}{{}^{2n}C_2} + \frac{1}{{}^{2n}C_3} - \dots + (-1)^n \frac{1}{{}^{2n}C_{n-1}} \right] + (-1)^{n-1} \cdot \frac{n}{{}^{2n}C_n}$$

$$\begin{aligned}
&= 2n \left\{ \frac{2n+1}{2(n+1)} \left(\frac{1}{{}^{2n+1}C_1} + \frac{1}{{}^{2n+1}C_2} \right) - \frac{2n+1}{2(n+1)} \left(\frac{1}{{}^{2n+1}C_2} + \frac{1}{{}^{2n+1}C_3} \right) + \frac{2n+1}{2n+1} \left(\frac{1}{{}^{2n+1}C_3} + \frac{1}{{}^{2n+1}C_4} \right) \dots \right. \\
&\quad \left. \dots + (-1)^n \cdot \frac{2n+1}{2(n+1)} \left(\frac{1}{{}^{2n+1}C_{n-1}} + \frac{1}{{}^{2n+1}C_n} \right) \right\} + (-1)^{n-1} \cdot \frac{n}{{}^{2n}C_n} \\
&= 2n \frac{(2n+1)}{2(n+1)} \left[\left(\frac{1}{{}^{2n+1}C_1} + \frac{1}{{}^{2n+1}C_2} \right) - \left(\frac{1}{{}^{2n+1}C_2} + \frac{1}{{}^{2n+1}C_3} \right) + \left(\frac{1}{{}^{2n+1}C_3} + \frac{1}{{}^{2n+1}C_4} \right) - \dots \right. \\
&\quad \left. \dots + (-1)^n \left(\frac{1}{{}^{2n+1}C_{n-1}} + \frac{1}{{}^{2n+1}C_n} \right) \right] + (-1)^{n-1} \cdot \frac{n}{{}^{2n}C_n} \\
&= 2n \frac{(2n+1)}{2(n+1)} \cdot \left[\frac{1}{{}^{2n+1}C_1} + (-1)^n \frac{1}{{}^{2n+1}C_n} \right] + (-1)^{n-1} \cdot \frac{n}{{}^{2n}C_n} \\
&= \frac{n}{n+1} + (-1)^n \cdot \frac{n! \cdot n!}{(2n)!} \cdot n + (-1)^{n-1} \cdot \frac{n}{{}^{2n}C_n} \\
&= \frac{n}{n+1}
\end{aligned}$$

7. For any natural number n , the number A given by $A = 2903^n - 803^n - 464^n + 261^n$ is divisible

- (1) 7 (2) 271 (3) 1897 (4) 13279

Sol. Answer (1, 2, 3)

$$A = (2903^n - 803^n) - (464^n - 261^n).$$

Now $a^n - b^n$ is always divisible $a - b$, therefore $2903^n - 803^n$ is divisible by $(2903 - 803) = 2100$ and $(464^n - 261^n)$ is divisible by $(464 - 261)$, i.e., 203. Since 2100 and 203 are both divisible by 7, it follows that A is divisible by 7.

$$\text{Again let us write } A = (2903^n - 464^n) - (803^n - 261^n).$$

Now $(2903^n - 464^n)$ is divisible by $(2903 - 464)$ i.e., 2439, which is a multiple of 271, so that $(2903^n - 464^n)$ is divisible by 271.

Also, $803^n - 261^n$ is divisible by $803 - 261$, i.e., 542 which is a multiple of 271, so that $803^n - 261^n$ is a multiple of 271.

Since $2903^n - 464^n$ and $803^n - 261^n$ are both multiples of 271, therefore of 271. Since A is a multiple of both 271 and 7 it follows that A is a multiple of 271×7 i.e., 1897.

8. The tens digit of $(81)^{100}(121)^{100} - 1$ is

- (1) 1 (2) 0 (3) 9 (4) 8

Sol. Answer (2)

$$\begin{aligned} & (81)^{100} (121)^{100} - 1 \\ &= (81 \times 121)^{100} - 1 \\ &= (99)^{200} - 1 = (100 - 1)^{200} - 1 \end{aligned}$$

Clearly the digits at units and tens place is 0.

9. Prove the equality

$$1^2 + 2^2 + 3^2 + \dots + n^2 = {}^{n+1}C_2 + 2({}^nC_2 + {}^{n-1}C_2 + \dots + {}^2C_2)$$

Sol. Since ${}^kC_2 = \frac{k(k-1)}{2} = \frac{k^2}{2} - \frac{k}{2}$ we get

$$\text{Hence } 2 {}^kC_2 = k^2 - k$$

$$2 \sum_{k=2}^n {}^kC_2 = \sum_{k=2}^n k^2 - \sum_{k=2}^n k \text{ which gives the required result.}$$

10. If S be such that $S = {}^nC_1 - 3 {}^nC_3 + 3^2 \cdot {}^nC_5 - 3^3 \cdot {}^nC_7 + \dots$; then S is equal to

$$(1) \quad (-1)^n \cdot \frac{2^n}{\sqrt{3}} \sin \frac{2n\pi}{3}$$

$$(2) \quad (-1)^{n+1} \cdot \frac{2^n}{\sqrt{3}} \sin \frac{2n\pi}{3}$$

$$(3) \quad (-1)^{n+1} \cdot \frac{2^n}{\sqrt{3}} \cos \frac{2n\pi}{3}$$

$$(4) \quad (-1)^n \cdot \frac{2^n}{\sqrt{3}} \cos \frac{2n\pi}{3}$$

Sol. Answer (2)

$$\omega^n = \left(-\frac{1}{2} + \frac{i\sqrt{3}}{2} \right)^n = \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right)^n = \cos \frac{2n\pi}{3} + i \sin \frac{2n\pi}{3}$$

$$\text{Now } \left(-\frac{1}{2} + \frac{i\sqrt{3}}{2} \right)^n = \frac{(-1)^n}{2^n} (1 - i\sqrt{3})^n$$

$$= \frac{(-1)^n}{2^n} \left[1 + {}^nC_1(-i\sqrt{3}) + {}^nC_2(-i\sqrt{3})^2 + {}^nC_3(-i\sqrt{3})^3 + \dots \right]$$

$$= \frac{(-1)^n}{2^n} \left[(1 - 3 {}^nC_2 + \dots) - i\sqrt{3}({}^nC_1 - 3 {}^nC_3 + 3^2 {}^nC_5 - 3^3 {}^nC_7 + \dots) \right]$$

Equating the real and imaginary parts, we get

$$S = (-1)^{n+1} \frac{2^n}{\sqrt{3}} \sin \frac{2n\pi}{3}$$

11. Show that $3^{2008} + 4^{2009}$ can be written as a product of two positive integers each of which is larger than 2009^{182} .

Sol. We have $3^{2008} + 4^{2009}$

$$= 3^{2008} + 2^{4018}$$

$$= (3^{1004} + 2^{2009})^2 - 2 \cdot 3^{1004} \cdot 2^{2009}$$

$$= (3^{1004} + 2^{2009})^2 - 3^{1004} \cdot 2^{2010}$$

$$= (3^{1004} + 2^{2009})^2 - (3^{502} \cdot 2^{1005})^2$$

$$= (3^{1004} + 2^{2009} - 3^{502} \cdot 2^{1005})(3^{1004} + 2^{2009} + 3^{502} \cdot 2^{1005})$$

$$\text{Now } 2^{11} > 2009$$

$$(2^{11})^{182} > (2009)^{182}$$

So, even the smaller factor is greater than $(2009)^{182}$.

